AN IRRATIONALITY MEASURE FOR CATALAN'S CONSTANT

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ABSTRACT. Let $G = \sum_{j\geq 0}(-1)^j(2j+1)^{-2}$ be Catalan's constant. We prove that $|G - a/b| > b^{-410}$ for every integer a and every sufficiently large integer b. Thus the irrationality exponent of G is at most 410. The proof builds on OpenAI's proof that G is irrational. That proof compares two bounds for a determinant whose entries are linear forms in 1 and G. The lower bound comes from the denominators, and the upper bound from an integral representation. We improve both sides and add the uniformity a measure needs. At the finite places we bound the rank of the reduction modulo p of the matrix of rational parts. This lowers the denominator at every prime in an explicit window. At the real place we bound the determinant of the row functions by a signed estimate. A Pick matrix argument reduces it to the block of the negative nodes. Heine's formula writes that block as a ratio of two log-gas integrals, and both are bounded by logarithmic energy. A Gauss quadrature formula extends the bound to an exponentially small neighbourhood of G, at a cost paid column by column. All steps are formalized in the Lean proof assistant, with only the standard axioms.

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1. Introduction

1.1. **The result.** Catalan's constant

$$G = \sum_{j=0}^{\infty} \frac{(-1)^j}{(2j+1)^2} = L(2, \chi_{-4}) = 0.91596\,55941\,77219\ldots$$

is the value at $s = 2$ of the Dirichlet L-function of the nontrivial character modulo 4. Its arithmetic has long been studied by analogy with $\zeta(2)$ and $\zeta(3)$. Until recently it was not known to be irrational. In October 2026 OpenAI [26] proved that G is irrational and verified the proof in Lean.

The *irrationality exponent* $\mu(\xi)$ of a real number ξ is the infimum of the real numbers μ such that $|\xi - a/b| > b^{-\mu}$ for all integers a and all sufficiently large integers b. Every irrational number has $\mu(\xi) \geq 2$, and almost every real number has $\mu(\xi) = 2$. An *irrationality measure* is an explicit upper bound for $\mu(\xi)$.

Theorem 1.1. *For every integer a and every sufficiently large integer b,*

$$\left| G - \frac{a}{b} \right| > b^{-410}.$$

In particular $\mu(G) \leq 410$.

The proof uses the prime number theorem. The theorem and every result it depends on have been verified in the Lean 4 proof assistant on top of OpenAI's formalization. The formal proof has no unproved steps and uses only the standard axioms (Section 9). The use of AI in this work is disclosed at the end of the paper.

1.2. **Background.** Irrationality proofs for values of L-functions at integers almost always follow the pattern of Apéry's proof for $\zeta(3)$ [1]. One constructs integer linear forms $q_n\xi - p_n$ that tend to zero, with $q_n \neq 0$, faster than the growth of their coefficients would allow for a rational ξ. If $|q_n\xi - p_n| = e^{-\alpha n + o(n)}$ and $|q_n| = e^{\beta n + o(n)}$, then ξ is irrational as soon as $\alpha > 0$, and $\mu(\xi) \leq 1 + \beta/\alpha$. Beukers [2] gave the linear forms for $\zeta(2)$ and $\zeta(3)$ as integrals of Legendre-type polynomials. Hata [15] refined such integrals. He used systematically the primes that divide all the coefficients of the forms, an idea that goes back to Chudnovsky. Rhin and Viola [29, 30] found that families of integrals for $\zeta(2)$ and $\zeta(3)$ are permuted by a finite group. The invariance of the integrals lets one divide all their coefficients by a large common factor. This *arithmetic method* gave the sharpest measures for $\zeta(2)$ and $\zeta(3)$ for many years. Marcovecchio [25] applied it to $\log 2$. Zeilberger and Zudilin [39] obtained $\mu(\pi) \leq 7.1032\ldots$ from Salikhov's integrals [33] with an additional arithmetic saving.

For Catalan's constant the analogous linear forms exist but do not decay fast enough. Rivoal and Zudilin [31] proved that one of the numbers $\beta(2), \beta(4), \ldots, \beta(2m)$ is irrational for suitable m, where β is the Dirichlet beta function. This does not single out $G = \beta(2)$. Zudilin [41] found an

Apéry-like recurrence for G. Its solutions give linear forms that tend to zero, but too slowly to prove irrationality. Related constructions and their limits are described in [40, 18, 19, 44]. Eskandari, Murty and Nemoto [10] construct a mixed motive with period G and obtain linear forms in 1 and G from it. Eskandari [9] improves the approximation exponent of these forms. Leibtag et al. [22] found further Apéry-like approximations to G in conservative matrix fields. Numerically the best of them still has a negative approximation exponent.

OpenAI's argument [26] has a different shape. It uses a determinant Δ of size $48N$ instead of a sequence of linear forms. The entries of Δ are rational linear combinations of 1 and G, given by double integrals against the Cauchy kernel $1/(1-ts)$. If G were rational, the denominators of the entries would give a lower bound $|\Delta| \geq e^{-(\mathsf{A}+o(1))n^2}$ with $n = 48N$. An integral representation gives the upper bound $|\Delta| \leq e^{-(\mathsf{U}+o(1))n^2}$. It rests on Andréief's identity [12], familiar from random matrix theory, and an energy estimate shows $\mathsf{U} > \mathsf{A}$. The margin is small, about $1.3 \cdot 10^{-4}$. The nonvanishing of Δ along the primes $N = p$ is proved separately by reduction modulo p. Determinants of linear forms were used for irrationality before, in Zudilin's determinantal criterion with Hankel determinants [43].

The first irrationality measure for G was obtained by Jha [16], who proved $\mu(G) \leq 10^8$. Carreira [7] treated the five integer parameters of the construction as free. He stated conditions under which OpenAI's argument goes through for general parameters and obtained $\mu(G) < 2294$. He also formulated the best real-place constant of the method as an equilibrium problem for two measures. In [6] he records attempts to carry the method to $\beta(4)$. Bojnal [3] instantiated Carreira's general theorem at the parameters $(36, 6, 2, 3, 2)$ and obtained $\mu(G) < 2401.74$. The search and certification were carried out with Claude Fable 5.1. Xu [38] reported $\mu(G) \leq 727.03$ for the parameters $(20, 3, 1, 2, 1)$, with an improved trial potential in the energy estimate. His certificate is numerical, in ball arithmetic, and is not formalized. Sun's preprint [34] had claimed a proof of the irrationality of G. Wachs [37] reports that its main estimate is not justified.

Since September 2026 a series of new results in this area has appeared, several of them obtained with AI systems. OpenAI proved $\mu(\pi) = 2$, with a Lean formalization [27]. Liu, Jiang and Zhang proved $\mu(\log r) = 2$ for every rational $r > 0$ with $r \neq 1$ [24]. Giraudin lowered the bound for $\mu(\pi^2)$ to 5.0193784 with two hypergeometric constructions of Zudilin [42] at new parameters [13]. He also lowered the bound for $\mu(\zeta(3))$ to 5.5138800 [14]. Leibtag, Barkan, Kalisch, Weinbaum, Shalyt and Kaminer [22] showed that the classical constructions are one-dimensional trajectories in conservative matrix fields. Searching over other trajectories, they obtained $\mu(\pi^2) < 4.62675$ and $\mu(\zeta(3)) < 5.4197$. Prellberg [28] proved that $\mathrm{Li}_2(-1/4)$, $\mathrm{Li}_2(1/5)$ and $\mathrm{Li}_2(-1/3)$ are irrational. Fauzan announced a proof that $\zeta(5)$ is irrational,

using Hankel determinants [11]. Lai combined this method with simultaneous approximations to show that 1, $\zeta(3)$ and $\zeta(4)$ are linearly independent over $\mathbb{Q}$ [20]. For $L(2,\chi_{-3})$, Calegari, Dimitrov and Tang proved that 1, π^2 and $L(2,\chi_{-3})$ are linearly independent over $\mathbb{Q}$ [4]. Their effective holonomy bounds give $\mu(L(2,\chi_{-3})) \le 24781$ [5, Thm. 5.1].

1.3. **The companion paper.** Catalan's constant is a difference of two values of the trigamma function ψ'. For $0 < s < t \le 1$ put

$$\Psi(s,t) = \psi'(s) - \psi'(t) = \sum_{n=0}^{\infty} \Big(\frac{1}{(n+s)^2} - \frac{1}{(n+t)^2} \Big).$$

Then $G = \frac{1}{16}\Psi(\frac{1}{4},\frac{3}{4})$, and also $L(2,\chi_{-3}) = \frac{1}{9}\Psi(\frac{1}{3},\frac{2}{3})$ and $\pi^2 = 3\Psi(\frac{1}{2},1)$. The irrationality exponent does not change under $\xi \mapsto r\xi + r'$ with $r, r' \in \mathbb{Q}$ and $r \neq 0$. So a measure for G is a measure for $\Psi(\frac{1}{4},\frac{3}{4})$. The companion paper [23] proves irrationality measures for the values $\Psi(s,1)$ with $s \in \{\frac{1}{4},\frac{3}{4},\frac{1}{3},\frac{2}{3},\frac{1}{2}\}$ and $s = \frac{k}{5}, \frac{k}{7}$, and for $\Psi(\frac{1}{3},\frac{2}{3})$. Among them are $\mu(G \pm 5\pi^2/48)$ and $\mu(L(2,\chi_{-3}))$. Its determinants are Löwner matrices of digamma differences. That method covers the pair $(\frac{1}{4},\frac{3}{4})$ as well, but its bound for this pair is too weak to give a measure. The present paper therefore refines OpenAI's construction instead. The two papers share their notation, their literature and the formalization repository.

1.4. **What a measure requires.** Write $\Delta_N(X)$ for the determinant regarded as a polynomial in the constant, so that $\Delta_N(G)$ is OpenAI's determinant. Let Q_N be the integer polynomial obtained from Δ_N by clearing denominators. Three properties of Q_N turn OpenAI's comparison into a measure.

(i) *Smallness on a neighbourhood of G.* The upper bound must hold not only at $X = G$. It must hold for all X with $|X - G| \le e^{-\rho N}$, and the exponent ρ enters the final bound linearly.

(ii) *Uniform nonvanishing.* One needs $Q_p(a/b) \neq 0$ for primes p with $p^2 \nmid b$, in place of $Q_p(G) \neq 0$.

(iii) *Arithmetic.* The integer $b^{\deg Q_N} Q_N(a/b)$ is nonzero, so it is at least 1 in absolute value. The smaller the denominator of Δ_N and its degree, the stronger the conclusion. The prime p must be chosen not to divide b, and a naive choice costs a factor.

Let d be the degree of Q_N divided by N and η the decay rate of Q_N normalised by N^2. Then the conclusion is $\mu(G) \le d\rho/\eta$ (Proposition 2.8). The work is to make d and ρ small and η large. Since η is a small difference of two large numbers, the gains in η matter most.

1.5. **New ideas.**

1. *All parameters at once.* The determinant has five integer parameters $(n_0, b_0, q_0, g_0, h_0)$, which we call a *design*. Every argument below is carried out for an arbitrary design, with the parameters as symbols. This covers

the degree, the denominators at all primes, the rank deficiencies, the nonvanishing and the estimates at the real place. A design is then data. We use $(58, 9, 3, 5, 3)$, so that $n = 58N$ and the degree is $53N$.

2. *Rank of the reduction modulo p, in closed form.* Let $\nu_N(p)$ be the dimension of the left kernel of the reduction modulo p of p^2 times the matrix of rational parts. At a prime p the valuation of Δ_N gains $\nu_N(p)$. We prove four lower bounds for $\nu_N(p)$ (Theorem 2.1). Each is a piecewise linear function of p, with coefficients linear in the five parameters, on its own window of primes. They come from antisymmetric combinations of rows, a reflection of Chebyshev polynomials modulo p, and the Frobenius relation $(1-X)^p = 1-X^p$. All four are proved from one digit formula for the entries modulo p.

3. *A counting inequality for the nullity.* The four proofs share one shape. A space $\mathcal{V}$ of row combinations is given, with a few linear conditions on $\mathcal{V}$, and the conditions force all column responses to vanish. We prove this once (Theorem 2.2). Then $\nu_N(p) \geq \dim \mathcal{V} - \operatorname{rank} \Phi$, where Φ is the map of conditions. A congruence from Lucas's theorem lets the inequality apply when the column filter is longer than the prime. Further families of kernel vectors are instances of it.

4. *A fixed rational subspace with stored kernel vectors.* In one form of the inequality the space is a tensor product of polynomial coordinates with a fixed subspace $V \subset \mathbb{Q}^s$ (Theorem 2.3). The required ranks are certified by integer vectors in the kernels of fixed matrices, stored once. This treats 34 windows of primes between $3N$ and $16.2N$.

5. *Paying for the perturbation column by column.* The smallness near G comes from a Gauss-type quadrature formula for the coefficient of G, with $53N$ nodes. A perturbed determinant is a sum of determinants in which some columns are replaced by point values. We bound each replaced column separately. A binomial bound for generalized Vandermonde determinants (Lemma 5.4) and a contour bound for the weight of each column (Lemma 5.1) give costs that peak at different columns. Their sum peaks lower than the sum of their peaks.

6. *A signed bound for the determinant of the row functions.* OpenAI's integrand contains a determinant $\det \mathsf{L}$ that the energy method bounds only by 6^n. We bound it on the configurations that matter (Section 6). A Pick matrix argument and a Schur complement reduce $|\det \mathsf{L}|$ to the block of the negative nodes, at a cost of 11 per positive node. The key input is a contour bound for a quadratic form on the model space of a Blaschke product (Lemma 6.3). Heine's formula and the Christoffel formula write the negative block as a ratio of two log-gas integrals. We bound the numerator from above by completing squares in Chebyshev moments, and the denominator from below by Jensen's inequality. An inequality for the logarithmic energy of signed measures (Theorem 6.8) keeps the combined bound concave. A separate argument treats positive nodes close to 1.

7. *Near-optimal configurations.* The signed bound needs hypotheses on the configuration. They hold on every configuration whose energy is within the gain of the maximum. On the other configurations OpenAI's bound is already strong enough. So the improved bound holds everywhere.

8. *Trial potentials with complex poles, interpolated in the number of fixed columns.* The energy estimates use trial sequences whose generating functions are logarithmic potentials of point charges near the edges of the support of the extremal measure. The signed bound uses three trials, at fractions 0, 1/200 and 1/16 of fixed columns. Between them the fields are affine in the fraction, and the remaining constant is an explicit quadratic.

9. *A criterion with a free lower end of the window.* In the choice of the prime, the window of candidates is $(V, U]$ with $U = \kappa \log b$ and $V = (\kappa - \omega) \log b$. We prove the criterion for every $\omega \in (0, 1]$ with the hypothesis $\kappa > 2\omega$ (Proposition 2.8). We use $\omega = 3/5$.

1.6. **The constants.** For the design $(58, 9, 3, 5, 3)$ the degree of Q_N is $d = 53N$. The perturbation exponent is $\rho = 296.586094$. The decay rate at the real place is $6431.1338\ldots$, and the denominators cost $\mathsf{A} = 6392.7880\ldots$, after a gain of $\int \kappa \geq 22.05325$ from the rank deficiencies. The signed bound contributes 8.29824 to the decay rate. The criterion uses $\eta = 38.33924$, $\omega = 3/5$, $\delta = 2.9 \cdot 10^{-7}$ and $\kappa = 410/\rho = 1.382397\ldots$ Its last condition reads $B > 409.99999\ldots$, and 410 is the least integer above it. Decimals are truncated. The constants of the formal proof are the exact rational numbers.

1.7. **Organization and notation.** Section 2 sets up designs and states the rank deficiencies, the valuations, the denominators, the bound at the real place and the criterion. Section 3 recalls OpenAI's construction for a design and proves the degree, the denominators and the nonvanishing. Section 4 proves the rank deficiencies. Section 5 reduces the bound near G to determinants with fixed columns and recalls the energy method. Section 6 proves the signed bound. Section 7 proves Theorem 1.1. Section 8 makes remarks on the method, and Section 9 describes the formalization.

We use the notation of OpenAI's paper [26] for the construction and refer to it for details. Citations of the form [26, Prop. 2.1] point to its numbered statements. We write log for the natural logarithm, v_p for the p-adic valuation and $\theta(x) = \sum_{p \leq x} \log p$ for Chebyshev's function. T_k and U_k are the Chebyshev polynomials of the first and second kind, with $U_{-1} = 0$, $T_{-k} = T_k$ and $U_{-k-1} = -U_{k-1}$. We write $[X^k]F$ or $F[k]$ for the coefficient of X^k in a polynomial F, $y_+ = \max(y, 0)$, and $\mathrm{rev}_E F(X) = X^E F(1/X)$ for the reversal of a polynomial of degree at most E.

2. Designs, rank deficiencies and the exponent

2.1. **Designs.** A *design* is a quintuple $(n_0, b_0, q_0, g_0, h_0)$ of integers with $n_0, b_0, q_0, h_0 \geq 1$, $g_0 \geq 0$ and $2g_0 < n_0$. For $N \geq 1$ we write

$$n = n_0 N, \quad b = b_0 N, \quad q = q_0 N, \quad g = g_0 N, \quad h = h_0 N,$$
$$L = n + b + q, \quad C = L + g, \quad H = C + h, \quad A = b + q + 2g,$$

and $d_0 = n_0 - g_0$. OpenAI's choice is the design $(48, 7, 4, 4, 2)$. We work throughout with

$$D = (58, 9, 3, 5, 3): \quad n = 58N, \quad b = 9N, \quad q = 3N, \quad g = 5N, \quad h = 3N,$$
$$L = 70N, \quad C = 75N, \quad H = 78N, \quad A = 22N,$$

and $d_0 = 53$. Section 3 recalls the construction of OpenAI's determinant for a design. It yields an $n \times n$ matrix $\mathcal{A}_N = \mathcal{A}_N^0 + G\mathcal{L}_N$ with rational matrices $\mathcal{A}_N^0$ and $\mathcal{L}_N$. The polynomial $\Delta_N(X) = \det(\mathcal{A}_N^0 + X\mathcal{L}_N)$ has degree at most $n - g$ (Lemma 3.1). The search that led to D is not part of the proof.

2.2. **Rank deficiencies.** For an odd prime p with $H < p^2$ the entries of $p^2\mathcal{A}_N^0$ are p-integral (Lemma 3.3). Let $\mathcal{A}_{N,p}^0$ be its reduction modulo p, and let $\nu_N(p)$ be the dimension of its left kernel, that is, n minus its rank. Write $\Lambda = g + q + h$ and $e = (p+1)/2 - q - h$.

Theorem 2.1. *Let D be a design, $N \geq 1$, and p an odd prime with $H < p^2$. Then $\nu_N(p)$ is at least each of the following quantities whenever the stated conditions hold. Negative values are replaced by 0.*

(a) (Theorem 4.1) If $2g < p$, $H \leq 4p$ and $D_T = \min(g, n - 1 - g - 2p, C - 3p) \geq 1$:

$$D_T - (n + q + g + h - 3p)_+.$$

(b) (Theorem 4.3) If $2h \leq p + 1$, $L \leq 3p \leq H$, $n + q + g + h \leq 3p$ and $2p \leq n + q$:

$$\min(n,\ 2p + 2g + 1) - g - 2p - 1.$$

(c) (Theorem 4.4) If $e \geq 0$, $n \leq 4p + 2g + 1$, $L \leq 6p$ and $b + q \leq p + \mu_0$, with $\mu_0 = \max(q, H + q - 6p)$:

$$\min\Big(\frac{p-1}{2} - g,\ n + b + q + h - \frac{11p+1}{2}\Big).$$

(d) (Theorem 4.6) If $e \geq 0$ and $n \leq 4p + 2g + 1$: $m_1 - m_0 - e$, with

$$m_0 = \max(b + q,\ L + \Lambda - 4p), \qquad m_1 = \min(p,\ L - 3p,\ b + q + 2g + 1).$$

These are the Lean theorems `t_rank`, `u_rank`, `u6_rank_closed` and `s_rank`, proved for every admissible design. For D they apply for $12N < p \leq 26N$, where they give most of the gain (Section 4.9).

Below $12N$ the closed formulas give nothing. Further kernel vectors come from a single counting inequality, of which the proofs of Theorem 2.1 are special cases. For $c \in \mathbb{F}_p^n$ and a raw column $j \in [b, L)$ let $R_c(j)$ be the

reduction modulo p of $p^2 \sum_r c_r R(r,j)$, with R as in (3.4). These are the responses of Lemma 3.3.

Theorem 2.2. *Let D be a design, $N \geq 1$, p an odd prime with $H < p^2$, and write $q = ap + r$ with $0 \leq r < p$. Let $\mathcal{V}$ be a finite-dimensional vector space over $\mathbb{F}_p$, $\iota\colon \mathcal{V} \to \mathbb{F}_p^n$ an injective linear map and Φ a linear map on $\mathcal{V}$. Suppose that for every $x \in \ker \Phi$ there is a polynomial $G_x \in \mathbb{F}_p[X]$ of degree less than r with*

$$\sum_{v=0}^{a} (-1)^{a-v} \binom{a}{v} R_{\iota(x)}(j + vp) = G_x(j) \qquad (b \leq j < L - ap).$$

Then $\nu_N(p) \geq \dim \mathcal{V} - \operatorname{rank} \Phi$.

For $q < p$ the hypothesis says that the conditions Φ force all raw responses to vanish. The case used for the counting is the *profile form*. There the space is a tensor product of a space of polynomial coordinates with a fixed rational subspace, and the conditions are jets composed with fixed linear maps.

Theorem 2.3. *Let $v_1, \dots, v_d \in \mathbb{Z}^s$ span a subspace $V \subset \mathbb{Q}^s$ of dimension d. For j in a finite set J let B_j be an integer matrix with s columns and $r_j \leq d$ an integer such that there are $d - r_j$ linearly independent vectors $k \in \mathbb{Z}^d$ with $B_j(\sum_a k_a v_a) = 0$. Let $\mathcal{E} \neq 0$ be the product of a nonzero $d \times d$ minor of $(v_1, \dots, v_d)$ and, for each j, of a nonzero maximal minor of these kernel vectors. Let $p \nmid \mathcal{E}$ be an odd prime with $H < p^2$, write $q = ap + r$ as above, let $V_p = \mathbb{F}_p^d$ be the space of coordinates of V modulo p, and let m and w_j ($j \in J$) be integers ≥ 0. Suppose that there are linear maps $\varphi_{j,k}\colon V_p^m \to V_p$ ($k < w_j$) and $\iota\colon V_p^m \to \mathbb{F}_p^n$ such that, on the joint kernel of the maps $B_j \circ \varphi_{j,k}$, the map ι is injective and satisfies the hypothesis of Theorem 2.2. Then*

$$\nu_N(p) \geq m\,d - \sum_{j \in J} w_j\, r_j.$$

In the applications m and the w_j are affine functions of N and p on a window of primes, while V and the B_j are fixed. So the bound is linear on the window. The data to be stored are the vectors v_a, the matrices B_j and the kernel vectors. For the design D these bounds, the closed formulas and the further families of Section 4 combine into the following.

Theorem 2.4. *For the design D there are 102 intervals $(a_i, b_i]$, with $16/7 = a_1 < b_1 \leq a_2 < \cdots < b_{102} = 53/2$, and a function κ, linear on each interval and 0 outside their union, with the following property. For all large N and every prime p with $a_i N < p \leq b_i N$ for some i,*

$$\nu_N(p) \geq N\kappa(p/N) - 2.$$

On each interval the bound is given by one of the families of Section 4. The area of κ is a rational number,

$$\int_0^\infty \kappa(x)\,dx = \frac{362355158746019386015965 98}{16426096632711159925781 25} = 22.05972\ldots \;\geq\; 22.05325.$$

The contributions by ranges of $x = p/N$ are as follows. The third column counts the intervals on which the bound is that of Theorem 2.3 with stored data.

range of x	intervals	with stored data	area
(16/7, 3]	11	0	0.06259
(3, 6]	46	23	0.25136
(6, 12]	27	9	0.71965
(12, 53/2]	18	2	21.02610
total	102	34	22.05972

The intervals in $(12, 53/2]$ are listed in Section 4.9. The full list is part of the formalization. On each interval one bound is used. The families come from different arguments. Except for one pair (Section 4.7) their kernels are not shown to be independent, so at a prime the bounds are not added.

2.3. **Valuations, denominators and the exponent.** The rank deficiencies enter through the following statement.

Proposition 2.5. *Let p be an odd prime with $H < p^2$ and let $z \in \mathbb{Q}$ have denominator prime to p. Then $p^2(\mathcal{A}_N^0 + z\mathcal{L}_N)$ has p-integral entries, its reduction modulo p equals $\mathcal{A}_{N,p}^0$, and*

$$v_p(\Delta_N(z)) \geq \nu_N(p) - 2n.$$

At the other primes there are the valuation bounds of OpenAI's method (Section 3). At 2 there is a bound from the expansion into minors. Together they give the denominators.

Proposition 2.6. *For the design D there are positive integers D_N with the following properties. $D_N\Delta_N(i) \in \mathbb{Z}$ for every integer i, and $q^{2n} \mid D_N$ for every prime $N < q \leq 2N$. For every $\varepsilon > 0$ and all large N*

$$\log D_N \leq (\mathsf{A} + \varepsilon)N^2,$$

$$\mathsf{A} = \frac{2673}{8}\log 2 + 116 + \frac{24269}{4} - \int_0^\infty \kappa(x)\,dx = 6392.7880\ldots$$

Here $\frac{2673}{8}\log 2$ is the contribution of the prime 2. The number $116 + \frac{24269}{4}$ is the area of the piecewise linear function that bounds the poles at the odd primes (Section 3). The last term is the gain of Theorem 2.4.

At the real place we prove the following in Sections 5 and 6. Put

$$\rho = n_0 c_2 + \Psi_0 = 296.586094, \qquad c_2 = 3.596893, \quad \Psi_0 = 87.9663. \tag{2.1}$$

Here c_2 is the energy cost of one fixed column and Ψ_0 bounds the logarithmic size of the weight and the binomial factor of one column, per unit of N.

Theorem 2.7. *Let $U_0 = U + 8.29824/n_0^2 = 2.2583256\ldots$, with $U = 2.255858836$ and $n_0 = 58$. For every $\varepsilon > 0$, all large N and all real x with $|x - G| \leq e^{-\rho N}$,*

$$|\Delta_N(x)| \leq \exp(-(n_0^2(U_0 - \tfrac{1}{2}\log 2) - \varepsilon)N^2).$$

The number $n_0^2(U - \frac{1}{2}\log 2) = 6422.8355\ldots$ is the decay rate that OpenAI's energy method gives for the design D. The signed bound adds 8.29824, so $n_0^2(U_0 - \frac{1}{2}\log 2) = 6431.1338\ldots$ Since $\mathsf{A} = 6392.7880\ldots$, the polynomial $Q_N = (d_0N)!\, D_N \Delta_N$ has integer coefficients, degree at most d_0N, and

$$|Q_N(x)| \le e^{-\eta N^2} \qquad (|x - G| \le e^{-\rho N}), \qquad \eta = 38.33924.$$

The exact difference is $38.34580\ldots$, and η is a rational number just below it. The remaining properties of Section 1.4 are the nonvanishing of $Q_p(a/b)$ for $p^2 \nmid b$ (Proposition 3.5) and the divisibility of the i-th coefficient of Q_N by q^{2i} for $N < q \le 2N$ (Lemma 3.4). They are used to choose the prime. The criterion is the following.

Proposition 2.8. *Let $\xi \in \mathbb{R}$, $d \in \mathbb{N}$, and let $\eta, \rho, \kappa, \delta, \omega > 0$ satisfy*

$$\omega \le 1, \quad 2\omega < \kappa, \quad \eta < 2d, \quad 1 < 2\big(\omega - \delta(2\kappa - \omega)\big), \quad d + 4d\delta\kappa < \eta\kappa.$$

Let $Q_p \in \mathbb{Z}[X]$ (p prime) satisfy the following for all large p. $\deg Q_p \le dp$. $Q_p(a/b) \ne 0$ whenever $a \in \mathbb{Z}$, $b \ge 1$ and $p^2 \nmid b$. For all primes $p < q \le 2p$ and $i \le dp$, q^{2i} divides the i-th coefficient of Q_p. Finally $|Q_p(x)| \le e^{-\eta p^2}$ for $|x - \xi| \le e^{-\rho p}$. Then

$$|\xi - a/b| > b^{-\rho\kappa}$$

for all integers a and all sufficiently large b.

The conditions say, in the order of the proof in Section 7, three things. The window of primes contains a prime not dividing b. The primes of the window above the chosen one are recycled. The final inequality is contradictory. For the design D one takes $d = 53$, $\omega = 3/5$, $\delta = 2.9 \cdot 10^{-7}$ and $\kappa = B/\rho$ with $B = 410$. The last condition then reads $B > x := d\rho/(\eta - 4d\delta) = 409.99999\ldots$

2.4. **A reading of the exponent.** For a fixed design the bound is $d\rho/\eta$ in the limit. Its three ingredients come from three different places. The degree $d = n - g$ comes from the construction. The perturbation exponent ρ comes from the quadrature and the energy cost of a column. The decay rate η is the difference of the real place and the denominators, 6431.1338 and 6392.7880. Both large numbers are improved here: the rank deficiencies lower the second by 22.05, and the signed bound raises the first by 8.29824. Without either, η would be 7.98775.

3. OpenAI's construction for a design

We recall the construction of [26] in the generality in which we use it, and we prove for an arbitrary design the degree bound, the bounds for the denominators at the finite places and the nonvanishing at the primes $N = p$. Carreira [7] checks that the argument of OpenAI goes through for designs satisfying a few conditions. We need the same for the finite places and, in addition, for the entries modulo p^2.

3.1. **The determinant.** Let $D = (n_0, b_0, q_0, g_0, h_0)$ be a design, with the notation of Section 2. For $0 \le r < n$ the row polynomials are

$$P_r(t) = (1-t)^h t^{C-1} T_{|r-g|}(1/t), \qquad \deg P_r < H, \tag{3.1}$$

together with companion polynomials D_r built from $U_{|r-g|-1}$ [26, (3)]. Thus P_r depends only on $|r-g|$. Put $f(t) = \sqrt{1-t^2}$ and $d\mu(t) = |t|\,dt/f(t)$ on $(-1,1)$, and

$$A_r(t) = P_r(t) - \frac{3}{2}\,\mathbf{1}_{\{t>0\}}\frac{f(t)}{t} D_r(t), \qquad \psi_k(s) = s^{b+k}(1-s)^q.$$

The determinant studied by OpenAI is that of the $n \times n$ matrix

$$\mathcal{A}_N(r,k) = \int_{-1}^{1}\int_{0}^{1} A_r(t)\,\frac{\psi_k(s)}{1-ts}\,ds\,d\mu(t) \qquad (0 \le r,k < n). \tag{3.2}$$

The factor $(1-s)^q$ in ψ_k amounts to a finite difference of order q applied to the columns. The entries are rational linear combinations of 1 and G: there are rational matrices $\mathcal{A}_N^0$ and $\mathcal{L}_N$ with

$$\mathcal{A}_N = \mathcal{A}_N^0 + G\,\mathcal{L}_N, \qquad \mathcal{L}_N(r,k) = \sum_{v=0}^{q} (-1)^v \binom{q}{v} \sum_i [t^i] P_r \cdot 4\,c_{(i-b-k-v)/2}, \tag{3.3}$$

where $c_l = 4^{-l}\binom{2l}{l}$ for integers $l \ge 0$ and $c_z = 0$ otherwise [26, Prop. 2.1 and (20)]. The rational part is

$$\begin{aligned} \mathcal{A}_N^0(r,k) &= \sum_{v=0}^{q} (-1)^v \binom{q}{v} R(r, b+k+v), \\ R(r,j) &= \sum_{i<H} P_r[i]\,M(i,j) - \frac{3}{2}\sum_{i<H} D_r[i]\,Z(i,j) \end{aligned} \tag{3.4}$$

with rational numbers $M(i,j)$ (moments of μ against the Cauchy kernel) and $Z(i,j)$, where $Z(i,i) = -\sum_{k\le i} k^{-2}$ and $Z(i,j) = (H_i - H_j)/(i-j)$ for $i \ne j$, $H_i = \sum_{k\le i} 1/k$ [26, Prop. 2.1]. The design enters (3.4) only through the integer coefficients $P_r[i], D_r[i]$ and the range of j, which is $[b, L)$. Put $\Delta_N(X) = \det(\mathcal{A}_N^0 + X\mathcal{L}_N) \in \mathbb{Q}[X]$, so that $\Delta_N(G) = \det \mathcal{A}_N$.

3.2. **The degree.**

Lemma 3.1. $\deg \Delta_N \le n - g$.

Proof. By (3.3) the row r of $\mathcal{L}_N$ depends only on P_r, and hence only on $|r-g|$. So the rows r and $2g-r$ of $\mathcal{L}_N$ agree for $0 \le r < g$, and $2g - r \le n-1$ because $2g < n$. Subtracting row r from row $2g-r$ for each $r < g$ does not change the determinant and makes g rows of the new coefficient of X vanish. A determinant $\det(B + XL')$ in which L' has $n-g$ nonzero rows has degree at most $n-g$ in X. □

3.3. **Rank deficiency and valuation.** We use repeatedly the following elementary fact.

Lemma 3.2. *Let M be an $n\times n$ matrix with entries in $\mathbb{Z}_{(p)}$ whose reduction modulo p has rank at most $n-e$. Then $v_p(\det M)\geq e$.*

Proof. Factor the reduction through $\mathbb{F}_p^{n-e}$ and lift the two factors to integral matrices U,V, so that $M=UV+pE$ with E integral at p. Expanding the determinant multilinearly in the columns, every term that uses more than $n-e$ columns of UV vanishes, and every other term has at least e factors p. □

3.4. **The digit formula.** The reduction of $p^2\mathcal{A}_N^0$ modulo p is described by the digits in base p of the coefficients of the row polynomials. This is the one place where the arithmetic of the moments enters. It is design-free. For an odd prime p and $c\in\mathbb{F}_p^n$ put

$$P_c=\sum_r c_r P_r,\qquad D_c=\sum_r c_r D_r,\qquad W=(1-X^2)^{(p-1)/2},$$

(reading the P_r, D_r as polynomials in X), and let $F[k]$ denote the coefficient of X^k.

Lemma 3.3 (digit formula)**.** *Let p be an odd prime with $H<p^2$, let $0\leq j<H$, and write $j=Jp+\ell$ with $0\leq\ell<p$. Then $p^2R(r,j)$ is p-integral for every row r, and for every $c\in\mathbb{Z}^n$*

$$p^2\sum_r c_r R(r,j)\equiv\sum_{u\geq 0}\Big(\alpha_{u,J}\,(XWP_c)[up+\ell]+\beta_{u,J}\,D_c[up+\ell]\Big)\pmod p,$$

where $\alpha_{u,J}=M(u-1,J)$ and $\beta_{u,J}=-\frac{3}{2}Z(u,J)$ are p-integral numbers that do not depend on c, on ℓ or on the design.

Proof. Two facts about the numbers M and Z are needed, for $0\leq i,j<p^2$. Write $j=Jp+\ell$ and let $\varrho\in[0,p)$ be the residue of $j-i-1$ modulo p, $u=(i+1+\varrho-\ell)/p$. First, $p^2M(i,j)$ is p-integral and $p^2M(i,j)\equiv W[\varrho]\,M(u-1,J)\pmod p$. This is the digit reduction of the moments in [26, §3], obtained from the p-adic analysis of their recursions, and it does not involve a design. Second, $p^2Z(i,j)$ is p-integral and is congruent to $Z(\lfloor i/p\rfloor,J)$ if $i\equiv j\pmod p$ and to 0 otherwise. For the second, write $i'=\lfloor i/p\rfloor$. Since $i<p^2$, $H_i=p^{-1}H_{i'}+S_i$ and $H_i^{(2)}=p^{-2}H_{i'}^{(2)}+S_i'$ with S_i,S_i' p-integral, which gives the case $i=j$. If $i\not\equiv j$, then $p^2Z(i,j)=p\cdot pZ(i,j)\equiv 0$. If $i\equiv j$, $i\neq j$, then $i-j=(i'-J)p$, and the integers $k\in(j,i]$ prime to p form $i'-J$ complete systems of residues modulo p, so that $\sum_k 1/k\equiv(i'-J)\sum_{r=1}^{p-1}1/r\equiv 0\pmod p$. Hence $H_i-H_j\equiv(H_{i'}-H_J)/p$ and $p^2Z(i,j)\equiv Z(i',J)$.

Now use (3.4). By the first fact the part $\sum_i F[i]\,p^2M(i,j)$ with $F=P_c$ is $\sum_i F[i]\,W[\varrho(i)]\,M(u(i)-1,J)$. For fixed u the indices with $u(i)=u$ are those for which $\varrho=up+\ell-1-i$ lies in $[0,p-1]$, and W has degree $p-1$, so the sum over these i is $(XFW)[up+\ell]$. This gives the α-terms. By the second fact

the part with D_c is $\sum_{i\equiv\ell} D_c[i]\, Z(\lfloor i/p\rfloor, J) = \sum_u Z(u,J) D_c[up+\ell]$, which gives the β-terms. The p-integrality of $M(u-1,J)$ and $Z(u,J)$ for $u, J < p$ is part of the first and second facts. □

For the filtered columns we use the following form. Let $0 \le k < n$, $s = b + k$, $J = \lfloor s/p \rfloor$ and $\mu = (s \bmod p) + q$. If $\mu < p$, the $q+1$ raw columns $s, \dots, s+q$ that enter column k have the same digit J and residues $(s \bmod p) + v < p$. Since $\sum_{v=0}^{q} (-1)^v \binom{q}{v} F[t+v] = [X^{t+q}]\big((X-1)^q F\big)$ for every polynomial F, Lemma 3.3 gives

$$p^2 \sum_r c_r \mathcal{A}^0_N(r,k) \equiv \sum_{u\ge 0} \Big(\alpha_{u,J}\, Q_c[up+\mu] + \beta_{u,J}\, \widehat{D}_c[up+\mu] \Big), \tag{3.5}$$

where $Q_c = (X-1)^q\, XW P_c$ and $\widehat{D}_c = (X-1)^q D_c$. A combination c whose responses vanish on all raw columns in $[b, L)$ lies in the left kernel of $\mathcal{A}^0_{N,p}$, and a combination whose responses on the columns outside a set vanish and which kills the set by (3.5) does as well.

3.5. **Integrality and the valuation at p.**

Proof of Proposition 2.5. By Lemma 3.3 the entries of $p^2\mathcal{A}^0_N$ are p-integral, since the filter is an integer combination of raw columns with $j < L \le H$. The entries of $\mathcal{L}_N$ are p-integral for odd p, since $P_r[i] \in \mathbb{Z}$ and $c_l \in \mathbb{Z}[1/2]$, so $p^2\mathcal{L}_N \equiv 0 \pmod{p^2}$ and, for z with denominator prime to p, $p^2(\mathcal{A}^0_N + z\mathcal{L}_N)$ has p-integral entries with the same reduction as $p^2\mathcal{A}^0_N$. By Lemma 3.2 the determinant of the latter has valuation at least $\nu_N(p)$, and $\det\big(p^2(\mathcal{A}^0_N + z\mathcal{L}_N)\big) = p^{2n}\Delta_N(z)$. □

3.6. **The denominators.**

Let p be an odd prime with $H < p^2$ and $p \le H$. The method of [26, §3], applied to a design, bounds from below the valuation of $\Delta_N(z)$ for every z with denominator prime to p by $-\ell_N(p)$, where $\ell_N(p)$ counts the columns whose entries have a pole at p, once the coefficients of the row polynomials are grouped by their digits in base p and the column filter is taken into account. It is given by two formulas, in which all differences are truncated at 0. For $2p \le H$,

$$\ell_N(p) = 2n - \Big(\min(p,\ L-p,\ A) - \max(b,\ H-2p) - q \Big),$$

and for $H < 2p \le 2H$, with $R = (L-b) - (\min(A, L-p) - b) - (\min(p, L) - \max(b, H-p))$, $P' = \min(A, L-p) - b$ and $E' = \min(b, H-p) - (L-p)$,

$$\ell_N(p) = \min\big(2n,\ n+R,\ 2R + P' + E'\big).$$

These are the exponents of the generic valuation theorems in the formalization. For the primes that interest us they are the same as in OpenAI's paper for its design. For the design D the function $\ell_N(p) = Nf(p/N)$ is piecewise

linear in $x = p/N$ ($\lfloor\cdot\rfloor$ rounding aside):

$$f(x) = \begin{cases} 116 & x \le 59/2, \\ 175 - 2x & 59/2 \le x \le 69/2, \\ 106 & 69/2 \le x \le 39, \\ 184 - 2x & 39 \le x \le 81/2, \\ 265 - 4x & 81/2 \le x \le 48, \\ 217 - 3x & 48 \le x \le 69, \\ 148 - 2x & 69 \le x \le 70, \\ 78 - x & 70 \le x \le 78, \end{cases}$$

and $f = 0$ for $x \ge 78$, since for $p > H$ the matrix $\mathcal{A}_N^0 + z\mathcal{L}_N$ is p-integral. Its integral over $[1, 78]$ is $24269/4$, and $f = 116 = 2 \cdot 58$ on $[0, 1]$. Note that $f(x) = 2n_0$ for $x \le 59/2$, which contains all the primes in $(N, 2N]$ and, in particular, the whole range $(16/7, 53/2]$ of the rank deficiencies of Theorem 2.4. There the bound of Proposition 2.5 improves $-\ell_N(p) = -2n$ to $\nu_N(p) - 2n$.

The primes $p \le \sqrt{H}$ are not covered by these bounds. For them we use the cruder bound $6n \log_p H$ for the exponent, which contributes $O(\sqrt{N}\, N \log N) = o(N^2)$ in total.

3.7. **The prime 2.** At $p = 2$ the entries are not integral and the bound comes from the expansion of the determinant into minors, in the manner of [26, §3]. Its exponent is governed by a quadratic form in the numbers m, ℓ of columns of two exceptional types, which is

$$\tfrac{3}{2}m^2 + \tfrac{1}{2}\ell^2 - (H + C - b)\, m - C\,\ell + Cn,$$

and the following identity shows that it is at least $-\mathsf{A}_2$ whenever $m, \ell \ge 0$, $m + \ell \le n$ and $C \ge n$, where

$$\mathsf{A}_2 = \frac{(n + H - b)^2}{8} - \frac{n^2}{2}. \tag{3.6}$$

With $s = n - m - l$ and $m_* = (n + H - b)/4$,

$$\tfrac{3}{2}m^2 + \tfrac{1}{2}\ell^2 - (H + C - b)m - C\ell + Cn + \mathsf{A}_2 = 2(m - m_*)^2 + s\,(C - n + m) + \tfrac{1}{2}s^2 \ge 0.$$

This gives a 2-adic bound $v_2(\Delta_N(i)) \ge -\mathsf{A}_2 - O(N \log N)$, uniform in the integer i. For D, $\mathsf{A}_2 = (58 + 78 - 9)^2/8 - 58^2/2 = 2673/8$ in units of N^2.

3.8. **The common denominator.**

Proof of Proposition 2.6. Let $\kappa_N(p)$ be the largest of the lower bounds for $\nu_N(p)$ of Section 4 that apply at p (and 0 if none does), for primes $p \le H$ with $H < p^2$, and put $e_N(p) = \min(\ell_N(p),\, 2n - \kappa_N(p))$. For the primes with $p^2 \le H$ use the crude bound above, for $p = 2$ the 2-adic bound, and let D_N be the product of $p^{e_N(p)}$ over $p \le H$. By Proposition 2.5 and the bounds of this section, $v_p(\Delta_N(i)) \ge -e_N(p)$ for every integer $i \ge 0$ and every p (for $p > H$ the matrix is p-integral), so $D_N \Delta_N(i) \in \mathbb{Z}$. A prime $q \in (N, 2N]$ has

$\kappa_N(q) = 0$, since no window contains it, and $\ell_N(q) = 2n$, so $q^{2n} \mid D_N$ (here $N \geq 79$ guarantees $q > \sqrt{H}$).

For the size of D_N, the sum of $\ell_N(p) \log p$ over the odd primes is, by the prime number theorem in the weighted form

$$\frac{1}{N} \sum_{p} F(p/N) \log p \longrightarrow \int_0^\infty F(x)\, dx \qquad (N \to \infty), \tag{3.7}$$

for bounded, Riemann integrable, compactly supported F, equal to $N^2 \int_0^{78} f(x)\, dx + o(N^2)$ with $\int_0^{78} f = 116 + 24269/4$. The primes in the intervals of Theorem 2.4 lie in $(16N/7, 53N/2]$ and have $\ell_N(p) = 116N$, so their exponent is at most $116N - N\kappa(p/N) + 2$. The sum of $(N\kappa(p/N) - 2) \log p$ over these primes is $N^2 \int \kappa + o(N^2)$ by (3.7). The 2-adic part contributes $\mathsf{A}_2 \log 2\, N^2 + o(N^2)$. □

The polynomial Δ_N has degree at most $n - g = d_0 N$ and takes values in $D_N^{-1}\mathbb{Z}$ at the integers. An integer-valued polynomial of degree m has coefficients in $\frac{1}{m!}\mathbb{Z}$ (Newton's interpolation formula), so

$$Q_N = (d_0 N)!\, D_N\, \Delta_N(X) \in \mathbb{Z}[X], \qquad \deg Q_N \leq d_0 N, \tag{3.8}$$

and $\log(d_0 N)! = o(N^2)$. In this way the gain from the rank deficiencies enters through the integer polynomial itself, and not through a separate common factor of its coefficients.

Lemma 3.4. *For all large N and every prime $N < q \leq 2N$, the j-th coefficient of Q_N is divisible by q^{2j}.*

Proof. For these primes the matrices $q^2 \mathcal{A}_N^0$ and $\mathcal{L}_N$ are q-integral by Lemma 3.3, and q^{2n} divides D_N. The identity

$$Q_N(X/q^2) = (d_0 N)!\, \frac{D_N}{q^{2n}} \det(q^2 \mathcal{A}_N^0 + X \mathcal{L}_N)$$

shows that $Q_N(X/q^2)$ has q-integral coefficients. Its j-th coefficient is $c_{N,j}/q^{2j}$. □

3.9. **Nonvanishing.**

Proposition 3.5. *Let D be a design and suppose that the three $n_0 \times n_0$ matrices B_σ, $\sigma \in \{0, 1, -1\}$, defined below, are invertible. Then for all large primes p and every rational z such that pz has denominator prime to p, $\Delta_p(z) \neq 0$. In particular $\Delta_p(a/b) \neq 0$ whenever $a \in \mathbb{Z}$, $b \geq 1$ and $p^2 \nmid b$.*

The matrices are those of OpenAI's reduction modulo p at $N = p$, with the design in place of 48: let B be the rational $(n_0 + 1) \times n_0$ matrix of the entries of the filtered rational part at $N = 1$ for the rows $0, \ldots, n_0$ (the row n_0 is an auxiliary row), and let B_σ have rows $B_r + \sigma B_{r+1}$, $0 \leq r < n_0$. For the design D, B has 59 rows and 58 columns.

Proof. Let $p > 4H_0$, with $H_0 = H/N = 78$, so that $H = H_0 p < p^2/4$, and let p be prime to the numerators and denominators of the entries of B and of the three determinants. In the construction at $N = p$ write the row index $r = pr_0 + i$ with $0 \le r_0 < n_0$ and $0 \le i < p$. Since $q = q_0 p$, $C = C_0 p$, $h = h_0 p$, $g = g_0 p$, the Frobenius relation $(1-t)^p \equiv 1 - t^p$ expresses the row polynomials modulo p through those of the design at $N = 1$ and the successor row $r_0 + 1$. This is where the auxiliary row enters. It is valid for rows on both sides of g because the polynomials are defined through T_{r-g} and U_{r-g-1} for signed indices. The filter satisfies $s^{b_0 p + k}(1 - s)^{q_0 p} = s^{\ell}(s^p)^{b_0 + k_0}(1 - s^p)^{q_0}$ in $\mathbb{F}_p[s]$, so the filtered columns of one residue class use the raw columns $b_0 + k_0, \dots, b_0 + k_0 + q_0$ of the design at $N = 1$ and never change the residue. By [26, Lemma 3.1 and eqs. (31), (35), (36)], which need only $H < p^2/4$, the reduction of $p^2 \mathcal{A}_p^0$ modulo p is then, after ordering the rows by (i, r_0) and the columns by (ℓ, k_0), the matrix $a^{\mathsf{T}} \otimes B_0 + b^{\mathsf{T}} \otimes B_1$, where a, b are the fixed $p \times p$ matrices of OpenAI's identity (51), with $\det a = 2^{-p(p-1)/2}$. The permutation matrix Π in this decomposition has the eigenvalue 0 once and ± 1 each $(p-1)/2$ times, and therefore

$$\det = (\det a)^{n_0} \, \det B_0 \, \det(B_0 + B_1)^{(p-1)/2} \det(B_0 - B_1)^{(p-1)/2} \neq 0$$

in $\mathbb{F}_p$. Hence $p^{2n} \Delta_p(0)$ is a p-adic unit. For z with pz of denominator prime to p, the difference $p^2 z \mathcal{L}_p = p\,(pz)\,\mathcal{L}_p$ is p-integral with zero reduction (the entries of $\mathcal{L}_p$ are p-integral), so the reduction of $p^2(\mathcal{A}_p^0 + z\mathcal{L}_p)$ is the same invertible matrix and $v_p(\Delta_p(z)) = -2n$. If $p^2 \nmid b$, then $p(a/b)$ has denominator prime to p. □

For the design D the invertibility of B_0, $B_0 + B_1$ and $B_0 - B_1$ is certified by exhibiting their inverses modulo the prime 1009, which exceeds $H_0 = 78$ so that every denominator of B is a unit there. The three determinants are then nonzero rational numbers. The computation modulo 1009 certifies the base matrices. The conclusion for all large primes is a theorem and not an extrapolation from finitely many primes. Nothing in the proof uses the proportions of the design except $2g < n$, and $H < p^2/4$.

4. Rank deficiencies modulo p

In this section we prove Theorems 2.1, 2.2, 2.3 and 2.4. We begin with the four closed formulas, whose proofs show the mechanism in its simplest form, then prove the counting inequality that underlies all of them, describe the further families obtained from it, and finally the certified data for finitely many windows and the resulting function κ for the design D. The principle is that of the arithmetic method of Hata and Rhin–Viola, in the form that suits a determinant: if the reduction modulo p of $p^2 \mathcal{A}_N^0$ has rank at most $n - e$, then by Lemma 3.2 and Proposition 2.5 the determinant $\Delta_N(z)$ gains a factor p^e at every z with denominator prime to p, in particular at every integer. The gain is in the denominator, and by (3.8) it is passed on to the coefficients of Q_N. We therefore need lower bounds for the dimension of the

left kernel of $\mathcal{A}^0_{N,p}$, that is, explicit row combinations $c \in \mathbb{F}_p^n$ with $c^{\mathsf{T}}\mathcal{A}^0_{N,p} = 0$. The tools are the same throughout: the identities for Chebyshev polynomials in characteristic p

$$\begin{aligned} T_p &= X^p, \qquad U_{p-1} = (X^2-1)^{(p-1)/2}, \\ T_{kp} &= T_k(X^p), \qquad U_{kp-1} = (X^2-1)^{(p-1)/2}\,U_{k-1}(X^p) \end{aligned} \tag{4.1}$$

in $\mathbb{F}_p[X]$ (the last two follow from $T_{kp} = T_k \circ T_p$ and $U_{kp-1} = (U_{k-1} \circ T_p)\,U_{p-1}$), the relation $(1-X)^p = 1 - X^p$, and the digit formula (3.5).

4.1. **Setting.** Fix a design, an integer $N \geq 1$ and an odd prime p with $H < p^2$. For $0 \leq r < n$ put $m_r = r - g$, so that $-g \leq m_r \leq n-1-g$, and $n-1-g \geq g$. In the notation of (3.1),

$$P_r = (1-X)^h \operatorname{rev}_{C-1} T_{m_r}, \qquad D_r = (1-X)^h \operatorname{rev}_{C-1} U_{m_r-1},$$

where $T_{-m} = T_m$ and $U_{-m-1} = -U_{m-1}$, so that $D_g = 0$. For $c \in \mathbb{F}_p^n$ write $S_c = \sum_r c_r \operatorname{rev}_{C-1} T_{m_r}$ and $U_c = \sum_r c_r \operatorname{rev}_{C-1} U_{m_r-1}$, so that $P_c = (1-X)^h S_c$ and $D_c = (1-X)^h U_c$. The lowest degree of $\operatorname{rev}_{C-1} T_m$ is at least $C-1-|m| \geq a := A = b+q+2g$, and that of $\operatorname{rev}_{C-1} U_{m-1}$ is at least $a+1$. Put

$$Q_c = (X-1)^q\, XW P_c, \qquad \widehat{D}_c = (X-1)^q D_c, \qquad W = (1-X^2)^{(p-1)/2},$$

so that, with $\Lambda = g+q+h$, Q_c has its coefficients in $[a+1,\, L+\Lambda+p-1]$ and $\widehat{D}_c$ in $[a+1,\, L+\Lambda-1]$. Recall that a vector c lies in the left kernel of $\mathcal{A}^0_{N,p}$ as soon as the right side of (3.5) vanishes for every filtered column k (for the columns with a carry among the $q+1$ raw columns one splits the sum over the raw columns at the carry. We shall not need this case).

4.2. **Antisymmetric combinations of rows.**

Theorem 4.1. *Let $2g < p$, $H \leq 4p$ and $D_T = \min(g,\, n-1-g-2p,\, C-3p) \geq 1$. Then $\nu_N(p) \geq D_T - (n+q+g+h-3p)_+$.*

The rows used are the differences of rows placed symmetrically about the three centres g, $g+p$ and $g+2p$.

Proof. For $1 \leq d \leq D_T$ and $k = 0, 1, 2$ let $\delta_k(d) = e_{g+kp+d} - e_{g+kp-d} \in \mathbb{F}_p^n$. These rows exist because $d \leq g$ and $g+2p+d \leq n-1$, and the $3D_T$ vectors $\delta_k(d)$ are linearly independent, since their supports are disjoint: the supports lie in the intervals $[kp, kp+2g]$ of row indices, which are disjoint because $2g < p$. The row $g+kp \pm d$ has $m = kp \pm d$. By $T_{a+b} - T_{a-b} = 2(X^2-1)U_{a-1}U_{b-1}$ and $U_{a+b-1} - U_{a-b-1} = 2T_aU_{b-1}$ together with (4.1),

$$\begin{aligned} T_{kp+d} - T_{kp-d} &= 2(X^2-1)^{(p+1)/2}\,U_{k-1}(X^p)\,U_{d-1}, \\ U_{kp+d-1} - U_{kp-d-1} &= 2\,T_k(X^p)\,U_{d-1}. \end{aligned}$$

Put $R_d = 2(1-X)^h X^{C-d}\operatorname{rev}_{d-1} U_{d-1}$, the value of $D_{\delta_0(d)}$. Reversing at degree $C-1$, and using that $\operatorname{rev}_{kp}(T_k(X^p)) = (\operatorname{rev}_k T_k)(X^p)$, one finds, with $T_0 = 1$, $T_1 = X$ and $T_2 = 2X^2 - 1$,

$$D_{\delta_1(d)} = X^{-p} R_d, \qquad D_{\delta_2(d)} = 2X^{-2p} R_d - R_d,$$

$$XWP_{\delta_0(d)} = 0, \qquad XWP_{\delta_1(d)} = X^{-p} R_d - X^p R_d,$$
$$XWP_{\delta_2(d)} = 2X^{-2p} R_d - 2R_d.$$

The polynomial R_d is supported in degrees $[C-d, H-1]$, which lies in $[3p, 4p)$ because $d \le C - 3p$ and $H \le 4p$. So at each residue ℓ it has a single base-p digit $\varrho_\ell(d)$, in position $u = 3$. For $x = (x_0, x_1, x_2) \in \mathbb{F}_p^3$ the response of the combination $\sum_k x_k \delta_k(d)$ at the raw column $Jp + \ell$ is therefore, by Lemma 3.3, $\varrho_\ell(d)\, f_J(x)$ with

$$f_J(x) = x_0\beta_{3,J} + x_1(\alpha_{2,J} - \alpha_{4,J} + \beta_{2,J}) + x_2(2\alpha_{1,J} - 2\alpha_{3,J} + 2\beta_{1,J} - \beta_{3,J}),$$

which depends neither on d nor on ℓ. The raw columns with $J \ge 3$ do not meet the support, since $3p + \ell \ge C - d \ge C - g = L$. Choose $x^* \ne 0$ with $f_1(x^*) = f_2(x^*) = 0$. This is two linear equations in three unknowns. For $c = \sum_d s_d \sum_k x_k^* \delta_k(d)$ the responses at the columns with $J = 1, 2$ vanish, and the columns with $J = 0$, which have $\ell \ge b$, respond by $f_0(x^*) \sum_d s_d \varrho_\ell(d)$, the coefficient of $X^{3p+\ell}$ in $\sum_d s_d R_d$. These are at most $(H - 3p - b)_+ = (n+q+g+h-3p)_+$ linear conditions on $s = (s_d)$. The map $s \mapsto c$ is injective, so the solutions form a space of dimension at least $D_T - (n+q+g+h-3p)_+$ in the left kernel. □

4.3. **A reflection of Chebyshev polynomials.** The first ingredient of the next theorem is a reflection formula modulo p.

Lemma 4.2. *Let p be an odd prime and $\nu = (p+1)/2$. In $\mathbb{F}_p[X]$, for every integer m,*

$$T_{p-m} = X^p T_m - (X^2 - 1)^\nu U_{m-1}.$$

Proof. Both sides satisfy the three-term recurrence in m, so it suffices to check $m = 0$ and $m = 1$. This needs (4.1). For $m = 0$ it is $T_p = X^p$. For $m = 1$ use $T_{p+1} = XT_p + (X^2 - 1)U_{p-1}$ and $T_{p-1} = 2XT_p - T_{p+1}$. □

Theorem 4.3. *Let $2h \le p+1$, $L \le 3p \le H$, $n + q + g + h \le 3p$ and $2p \le n+q$, and put $n^* = \min(n, 2p+2g+1)$. Then $\nu_N(p) \ge n^* - g - 2p - 1$.*

Proof. We restrict c to the rows $r < n^*$, so that $m_r \le n^* - 1 - g \le 2p + g$. Applying Lemma 4.2 with $m = m_r$ and reflecting at degree $p + C - 1$ gives, for every such row,

$$(1 - X^2)^\nu \operatorname{rev}_{C-1} U_{m-1} - X \operatorname{rev}_{C-1} T_m = -X^{p+1} \operatorname{rev}_{C-1} T_{p-m},$$

and the right side is divisible by $X^{p+C-|p-m|}$, hence by X^L since $|p - m| \le p + g = p + C - L$. Multiplying by $(1-X)^h$ and summing, for every c supported on these rows

$$X^L \mid (1 - X^2)^\nu D_c - XP_c. \tag{4.2}$$

Let V be the space of such c with $(1-X^2)^\nu \mid P_c$. Since $(1-X)^h$ divides every P_r, this is $\nu-h$ conditions at $X=1$ and ν conditions at $X=-1$, that is, at most $p+1-h$ linear conditions. For $c\in V$ write $P_c=(1-X^2)^\nu Y$ and $\widetilde{Q}=XY$. Since $(1-X^2)^p=1-X^{2p}$,

$$XWP_c=(1-X^{2p})\,\widetilde{Q}.$$

The polynomial P_c has no coefficients below X^{A^*}, $A^*=C-n^*+g$, and none from X^H on, so $\widetilde{Q}$ is supported in $[A^*+1,\,H-p-1]$. Dividing (4.2) by the unit $(1-X^2)^\nu$ modulo X^L gives $D_c[k]=\widetilde{Q}[k]$ for $k<L$.

Now let $H-3p\le\ell<L-2p$. The inequality $L-2p\le A^*+1$ is equivalent to $n^*\le 2p+2g+1$, and $L-2p\le p$ since $L\le 3p$. By the supports, $\widetilde{Q}[\ell]=0$, $\widetilde{Q}[2p+\ell]=0$ and $\widetilde{Q}[3p+\ell]=0$, and $D_c[3p+\ell]=0$ because $3p+\ell\ge H$. So with $\varkappa=\widetilde{Q}[p+\ell]$ the base-p digits of XWP_c at the residue ℓ are $(0,\varkappa,0,-\varkappa,0,\dots)$ and those of D_c are $(0,\varkappa,0,0,\dots)$. Hence, by Lemma 3.3, the response at every raw column $Jp+\ell$ with such an ℓ is $\varkappa$ times a constant depending on J only: on V these columns, which have $J\in\{0,1,2\}$, span a space of dimension at most one for each residue ℓ.

Put $\ell_0=H-3p$ and $\ell_1=L-2p$. The hypotheses give $\ell_0\le b\le\ell_1$. For $\ell_0\le\ell<b$ the raw columns $j\in[b,L)$ with this residue have $J\in\{1,2\}$, and for $b\le\ell<\ell_1$ they have $J\in\{0,1,2\}$. So the saving is $2\ell_1-\ell_0-b$, and the number of linear conditions needed to make all responses on the $n+q=L-b$ raw columns vanish on V is $(p+1-h)+(n+q)-(2\ell_1-\ell_0-b)$. Substituting $\ell_1=L-2p$, $\ell_0=H-3p$ this is $2p+1+g$. The space of solutions in the span of n^* rows has dimension at least $n^*-g-2p-1$. □

4.4. **A block of columns: the digits 1 to 5.**

Theorem 4.4. *Let $e=(p+1)/2-q-h\ge 0$, $n\le 4p+2g+1$, $L\le 6p$, and put $\mu_0=\max(q,H+q-6p)$, $\mu_1=L-5p$. Assume $b+q\le p+\mu_0$. Then $\nu_N(p)\ge\mu_1-\mu_0-e$, that is*

$$\nu_N(p)\ge\min\Big(\frac{p-1}{2}-g,\ n+b+q+h-\frac{11p+1}{2}\Big).$$

The first term is the minimum if $6p\le H$ and the second if $6p\ge H$.

Unlike Theorems 4.1 and 4.3, where the conditions are imposed on the rows, the argument concerns a block of columns. We first need an identity.

Lemma 4.5. *In $\mathbb{F}_p[X]$, for every integer k, $2X^p(X^2-1)^{(p-1)/2}T_k-(2X^{2p}-1)U_{k-1}=U_{2p-k-1}$. If $n\le 4p+2g+1$, then for every c, $X^L\mid 2XWS_c-(2-X^{2p})U_c$, and consequently*

$$2\,Q_c[i]=2\,\widehat{D}_c[i]-\widehat{D}_c[i-2p]\qquad(i<L).\tag{4.3}$$

Proof. For all integers m,k one has $U_{m-1}T_k-T_mU_{k-1}=U_{m-k-1}$, since both sides satisfy the same recurrence in k and agree for $k=0,1$. Take $m=2p$ and use (4.1): $T_{2p}=2X^{2p}-1$ and $U_{2p-1}=2T_pU_{p-1}=2X^p(X^2-1)^{(p-1)/2}$. Reflecting at degree $2p+C-1$, the first term becomes $2XW\,\mathrm{rev}_{C-1}T_k$, the

second $(2 - X^{2p})\operatorname{rev}_{C-1} U_{k-1}$, and the right side $X^{2p+C-1}U_{2p-k-1}(1/X)$. For $k = m_r \geq -g$ this is divisible by X^{C+k}, hence by X^L, if $2p-k-1 \geq 0$. If $2p-k-1 < 0$ it has degree $k-2p-1$ and is divisible by X^{4p+C-k}, hence by X^L when $k \leq 4p+g$, which holds because $k \leq n-1-g \leq 4p+g$. Summing with weights c_r and multiplying by $(X-1)^q(1-X)^h$ gives (4.3). □

Proof of Theorem 4.4. Let $M = [\mu_0, \mu_1)$ and $w = \mu_1 - \mu_0$. We may assume $w > e$. The block consists of the filtered columns $k(J,\mu) = Jp + \mu - (b+q)$ for $1 \leq J \leq 5$ and $\mu \in M$. These indices are non-negative by the hypothesis $b + q \leq p + \mu_0$, and smaller than n because $5p + \mu_1 - 1 = L - 1$. There are $5w$ of them. As $\mu < p$, formula (3.5) applies: the response of c at $k(J,\mu)$ is $\sum_u \alpha_{u,J} Q_c[up+\mu] + \beta_{u,J}\widehat{D}_c[up+\mu]$.

Supports. For $\mu \in M$ we have $p + \mu \leq p + \mu_1 - 1 = L - 4p - 1 \leq a$ by $n \leq 4p + 2g + 1$, so the digits $u = 0, 1$ of Q_c and $\widehat{D}_c$ vanish. From $\mu_0 \geq H + q - 6p = L + \Lambda - 6p$ we get that the digits $u \geq 6$ of $\widehat{D}_c$ and $u \geq 7$ of Q_c vanish. For $u \leq 5$ we have $up + \mu \leq L - 1$, so (4.3) applies.

From Q to $\widehat{D}$. For $u = 2, \ldots, 5$, (4.3) gives $\widehat{D}_c[up+\mu] = Q_c[up+\mu] + \frac{1}{2}\widehat{D}_c[(u-2)p+\mu]$, and for $u = 2, 3$ the last index is at most a, so the term vanishes. Hence if $Q_c[up+\mu] = 0$ for $u = 2, \ldots, 5$ and all $\mu \in M$, then by induction $\widehat{D}_c[up+\mu] = 0$ for $u = 2, \ldots, 5$, and all digits of $\widehat{D}_c$ in M vanish.

A relation from the Frobenius map. We have $Q_c = (-1)^q X(1-X)^{q+h+(p-1)/2}(1+X)^{(p-1)/2} S_c$, and $q + h + (p-1)/2 + e = p$. Since $(1-X)^p = 1 - X^p$,

$$(1-X)^e\, Q_c = (1 - X^p)\, Y_c, \qquad Y_c = (-1)^q X(1+X)^{(p-1)/2} S_c,$$

and Y_c is supported in $[a+1,\, C + (p-1)/2]$. Comparing the coefficients at $up + \mu$, subtracting the coefficient at $(u-1)p + \mu$, and summing over $u = 2, \ldots, 6$, the right side telescopes to $Y_c[6p+\mu] - Y_c[p+\mu]$, which vanishes for $\mu \geq \mu_0 + e$: the second term because $p + \mu \leq a$, the first because $6p + \mu \geq H + q + e = C + (p+1)/2 > \deg Y_c$. So, if $Q_c[up + \mu'] = 0$ for $u = 2, \ldots, 5$ and all $\mu' \in M$, then

$$\sum_{t=0}^{e} (-1)^t \binom{e}{t} Q_c[6p + \mu - t] = 0 \qquad (\mu_0 + e \leq \mu < \mu_1).$$

This is a recurrence with unit leading coefficient: if $Q_c[6p+\mu] = 0$ for the first e values $\mu = \mu_0, \ldots, \mu_0 + e - 1$, then it vanishes for all $\mu \in M$.

Counting. Impose on c that the $n - 5w$ filtered columns outside the block vanish, that $Q_c[up+\mu] = 0$ for $2 \leq u \leq 5$ and $\mu \in M$ ($4w$ conditions), and that $Q_c[6p+\mu] = 0$ for the first e values of μ. By the three steps above all the responses of the block columns vanish. These are at most $n - 5w + 4w + e = n - (w - e)$ linear conditions, so the left kernel has dimension at least $w - e$. Finally, $w - e$ equals $\mu_1 - q - e = n + b + q + h - (11p+1)/2$ if $\mu_0 = q$ and $(p-1)/2 - g$ if $\mu_0 = H + q - 6p$. □

4.5. **The second window.**

Theorem 4.6. *Let $e = (p+1)/2 - q - h \geq 0$ and $n \leq 4p + 2g + 1$, and put*

$$m_0 = \max(b+q,\ L+\Lambda-4p), \qquad m_1 = \min(p,\ L-3p,\ a+1),$$

where $a = b+q+2g$. Then $\nu_N(p) \geq m_1 - m_0 - e$.

Proof. Let $M = [m_0, m_1)$ and $w = m_1 - m_0 > e$. For $\mu \in M$ we have $\mu \leq a$, so the digit $u = 0$ of Q_c and $\widehat{D}_c$ vanishes. The inequality $\mu \geq L + \Lambda - 4p$ gives that the digits $u \geq 4$ of $\widehat{D}_c$ and $u \geq 5$ of Q_c vanish. Finally $3p+\mu < L$, so that (4.3) gives, with $D_i = \widehat{D}_c[ip+\mu]$ and $Q_i = Q_c[ip+\mu]$,

$$Q_1 = D_1, \qquad Q_2 = D_2, \qquad Q_3 = D_3 - \tfrac{1}{2}D_1.$$

So every response of the filtered matrix at a column with residue $\mu \in M$ is a combination of the four functionals $D_1(\mu), D_2(\mu), D_3(\mu), Q_4(\mu)$ of c. For $\mu \in M$ the columns $k = \mu - (b+q)$, $p + \mu - (b+q)$, $2p + \mu - (b+q)$ and $3p+\mu-(b+q)$ exist, since $b+q \leq \mu < p$ and $3p+\mu < L$. So there are at least $4w$ block columns, and their rows lie in the span F of the $4w$ functionals.

We find $w-e$ linear relations among these functionals. As in the proof of Theorem 4.4, $(1-X)^e Q_c = (1-X^p)Y_c$ with Y_c a polynomial. Put $\sigma(\ell) = \sum_{u \geq 0} Q_c[up+\ell]$ and sum the coefficients of this identity over the residue class of $\mu \geq e$ modulo p. The right side gives $\sum_u (Y_c[up+\mu] - Y_c[(u-1)p+\mu]) = 0$, since Y_c is a polynomial, and the left side gives $\sum_{t=0}^{e} (-1)^t \binom{e}{t} \sigma(\mu - t)$. For $\mu \in M$ with $\mu - e \geq m_0$ all $\mu - t$ lie in M, and there $\sigma = Q_1 + Q_2 + Q_3 + Q_4 = D_1 + D_2 + D_3 - \frac{1}{2}D_1 + Q_4$. So

$$\sum_{t=0}^{e} (-1)^t \binom{e}{t} (\tfrac{1}{2}D_1 + D_2 + D_3 + Q_4)(\mu - t) = 0 \qquad (m_0 + e \leq \mu < m_1),$$

and these $w-e$ relations are independent, since $Q_4(\mu)$ occurs with coefficient 1 in the one for μ and only with lower arguments in the others. Hence $\dim F \leq 4w - (w-e)$, and the rank of the matrix is at most $(n-4w)+3w+e = n - (w-e)$. □

The proof does not use a determinant or a lower bound for N. Unlike the proof of Theorem 4.4, the telescoping here is over all digits, so no vanishing of Y_c at the ends is needed.

4.6. **The counting inequality.** The four proofs above have one shape. A space of row combinations is chosen, a few linear conditions are imposed, and it is shown that the conditions make every response vanish. The dimension of the space minus the number of conditions is then a lower bound for the nullity. Theorem 2.2 isolates this argument, and allows in addition filters longer than the prime.

Proof of Theorem 2.2. By (3.4), the response of c at the filtered column k is $\sum_{v=0}^{q} (-1)^v \binom{q}{v} R_c(b+k+v)$, the value at T acting as the shift of the

polynomial $(1-T)^q$. Modulo p we have $(1-T)^p = 1-T^p$, hence

$$(1-T)^q = (1-T)^r(1-T^p)^a,$$

which is the congruence of Lucas's theorem for the binomial coefficients $\binom{q}{v}$. So the response at k equals $\sum_{u=0}^{r}(-1)^u\binom{r}{u}S_c(b+k+u)$, where $S_c(j) = \sum_{v=0}^{a}(-1)^v\binom{a}{v}R_c(j+vp)$. The arguments satisfy $b \le j$ and $j+ap \le b+n-1+q = L-1$. For $c = \iota(x)$ with $x \in \ker\Phi$ we have $S_c(j) = \pm G_x(j)$, and the difference of order r of a polynomial of degree less than r vanishes identically. So $\iota(\ker\Phi)$ lies in the left kernel of $\mathcal{A}^0_{N,p}$, and since ι is injective, $\nu_N(p) \ge \dim\ker\Phi = \dim\mathcal{V} - \operatorname{rank}\Phi$. □

If the conditions only force the responses on a set K of filtered columns to vanish, the same argument, with the remaining $n-|K|$ responses added to the conditions, gives $\nu_N(p) \ge \dim\mathcal{V} - \operatorname{rank}\Phi - (n-|K|)$. The proofs of Theorems 4.1–4.6 are of this form, with K the set of all filtered columns in Theorems 4.1 and 4.3 and a block of columns in Theorems 4.4 and 4.6.

Proof of Theorem 2.3. Since $p \nmid \mathcal{E}$, the reductions of $v_1,\dots,v_d$ are linearly independent, so that V_p is the reduction of V in the coordinates of this basis. The kernel vectors of B_j satisfy an integer identity, so their reductions lie in the kernel of $B_j|_{V_p}$, and they remain independent because their minor is a unit modulo p. Hence $\operatorname{rank}(B_j|_{V_p}) \le r_j$. The map $\Phi = (B_j\circ\varphi_{j,k})_{j,k}$ therefore has rank at most $\sum_j w_j r_j$, and its kernel has dimension at least $md - \sum_j w_j r_j$. Theorem 2.2, applied to $\ker\Phi$ with no further conditions, gives the bound. □

In the applications the row combinations are organized in *slot systems*. A slot system consists of data $x_1,\dots,x_D \in \mathbb{F}_p^K$, a linear map $x \mapsto c(x)$ from the data to row combinations, injective on a subspace of admissible data, fixed polynomials $\lambda_1,\dots,\lambda_D \in \mathbb{F}_p[X]$ of degree less than some $e_{\max}$, and two linear maps $\mathcal{T},\mathcal{G}$ from $\mathbb{F}_p^K$ to $\mathbb{F}_p[Y]$, such that for every admissible x the two row polynomials of $c(x)$ multiplied by $(X-1)^h$, respectively by $(X-1)^h(X^2-1)^{(p-1)/2}$, equal $\sum_d \lambda_d(X)\,\mathcal{T}(x_d)(X^p)$, respectively $\sum_d \lambda_d(X)\,\mathcal{G}(x_d)(X^p)$. This is the form in which (4.1) and the relation $(1-X)^p = 1-X^p$ express the row polynomials modulo p. By the digit formula, the response of $c(x)$ at the raw column $Jp+\ell$ is

$$\sum_{0\le e<e_{\max},\ p\mid H-1-\ell-e} \psi_{J,\,(H-1-\ell-e)/p}\Big(\sum_d \lambda_d[e]\,x_d\Big),$$

where $\lambda_d[e]$ is the coefficient of X^e and each $\psi_{J,s}$ is a linear form, a moment form in J and the digit s, determined by $\mathcal{T}$ and $\mathcal{G}$. For the stored subspaces the data are m polynomial coordinates with values in V, and the admissibility of the data is not imposed directly: the jets $\varphi_{j,k}$ are Taylor coefficients at $X=-1,0,1$, coefficients at ∞, and window rows, the map ι from V_p^m to the data is an arbitrary linear map, and only on the joint kernel of the

jets is it required to land in the admissible data, to be injective, and to give responses of the form needed in Theorem 2.2.

4.7. **Further families.** The following lower bounds for $\nu_N(p)$ are proved, each for every admissible design, by explicit choices of the space, the conditions and the columns in Theorem 2.2 or its variant above. Each is a piecewise linear function of p and the parameters, positive on windows described by linear inequalities. We describe them in words. The statements are those of the formalization, cited in Section 9.

(i) *Frobenius products.* Row combinations whose polynomial is the product of a fixed factor, a free polynomial P of bounded degree d_P in the Chebyshev variable, and a polynomial $\Lambda(X^p)$ with exponents in a window of K consecutive values. The conditions require Λ to annihilate the functionals seen by the columns, and the bound is $(d_P+1)(K-\#\text{seen})$. For D this gives $53N-2p-1$ on $(26N, 53N/2)$, where $2p+2 \le 53N$.

(ii) *A truncated form of Theorem 4.4.* Without the hypothesis $n \le 4p+2g+1$, the bound $\mu_1-\mu_0-e_+-(n-4p-2g-1)_+$ holds whenever $L \le 6p$ and $b+q \le p+\mu_0$. For D it gives $5p-59N+1$ on $(59N/5, 12N]$.

(iii) *A raw tail of the antisymmetric family.* With the four centres $g+kp$ $(0 \le k \le 3)$ and $D'_T = n-1-g-3p$, the count is $p-1-(2g+q+h)$ under $4p \le L$, $b \le p$, $D'_T \le g$, $D'_T \le (p-1)/2$, $3p+1+g \le n$ and $4p+b \le H$. For D this is $p-16N-1$ on $(16N, 118N/7]$.

(iv) *Two families of slot polynomials and their sum.* The first fixes a digit $s \le H/p$ and a range of blocks of rows, with slot polynomials divisible by a power of $X-1$. The second has slot polynomials $(X^p-1)P$ with P supported in a window of degrees. Their kernels are independent when both apply, under a separation condition, at the same window s or at the windows s and $s-1$, and the sum of the two bounds is a lower bound. This is the one place where two bounds are added.

(v) *Families with a constant profile on a cell.* Several families (the theorems called MT, M, Z, C and C′ in the formalization) use slot polynomials whose coordinates satisfy finitely many rational linear equations, the *cell* equations. A solution modulo p exists for every p prime to a fixed integer, the modulus of the cell, and the bound is the dimension of the slot space minus the number of conditions. In the families C and C′ the responses depend only on the residue of the column modulo p, so that the filter costs $(p-q)_+$ conditions. In C′, where $h \ge p$, one writes $(X-1)^h = (X^p-1)(X-1)^{h-p}$, and the factor $(X-1)^{h-p}$ costs $h-p$ more. The family C gives bounds on $(58N/17, 59N/17]$ and $(29N/8, 11N/3]$, and C′ on $(16N/7, 5N/2]$.

(vi) *Block decompositions for $h \ge p$ and $h < p$.* Writing a row index as $kp+d$ and $y = X^p$, the identities

$$(X-1)^h W T_{kp+d} = \big((X-1)T_k\big)(y)\, Q_1 + \big((X-1)(X^2-1)U_{k-1}\big)(y)\, Q_2$$

and its analogue for U_{kp+d-1}, with $Q_1 = (X-1)^{h-p} W T_d$ and $Q_2 = (X-1)^{h-p} U_{d-1}$, hold in $\mathbb{F}_p[X]$ for $h \ge p$. At a residue with a single live digit the

responses of all rows of a block are proportional, and the bound counts the blocks. For $h < p$ a difference of order $p - h$ of the coefficients plays the same role. For D these give bounds on $(59N/21, 3N]$, where $h \geq p$, and on $(3N, 16N/5]$.

4.8. **Stored data for finitely many windows.** On the remaining windows below $16.2N$ the bound is that of Theorem 2.3 with a stored subspace. For each of 34 windows we store integer vectors $v_1, \dots, v_d$ spanning V, a set of d pivot columns with the adjugate of the corresponding minor, and for each map B_j its integer matrix, the bound r_j, $d - r_j$ kernel vectors in the coordinates of V, and again a pivot set with an adjugate. A single Boolean function of these data, evaluated by the Lean kernel, checks the shapes, the identities $\mathrm{adj} \cdot v_{\mathrm{piv}} = \delta \cdot I$ and the products $B_j(\sum_a k_a v_a) = 0$ in integers. Its soundness theorem gives the hypotheses of Theorem 2.3 on V and the B_j, with $\mathcal{E}$ the product of the minors. No prime enters this check.

The hypotheses on the jets and on ι depend on N and p. A window is described by finitely many inequalities that are affine in N and $(p-1)/2$, and so are m and the w_j on each of finitely many pieces of the window. A second verified procedure checks on each piece that the numbers of jets and their orders at -1, 0, 1 and ∞ are nonnegative and fit the window, that $p > 59$, that the map ι is injective on the joint kernel and lands in the admissible data of a slot system, and that the response is given on the joint kernel by a polynomial in the column of degree less than q mod p (so that Theorem 2.2 applies), each required inequality in N and $(p-1)/2$ being derived from the window inequalities by sums, nonnegative integer multiples, and division of the variable part by a common divisor with rounding of the constant, that is, by Chvátal–Gomory cuts. Its soundness theorem gives the hypotheses of Theorem 2.3 at every odd prime $p > 59$ of the window with $p \nmid \mathcal{E}$. For large N all primes of the window are of this kind. For 33 of the 34 windows the hypotheses are established in this way, and for the remaining one by a direct proof of the same statement. The windows lie in $(3N, 16.2N)$, where $p > q$, so $a = 0$ in Theorem 2.2.

4.9. **The function κ for the design D.** For D, $L = 70N$, $H = 78N$, $a = 22N$, $\Lambda = 11N$ and $q + h = 6N$. On $(12N, 53N/2]$ the function κ of Theorem 2.4 is as follows, with $x = p/N$.

interval of x	$\kappa(x)$	bound
$(12, 13]$	$x/2 - 5$	Theorem 2.1(c), $(p-1)/2 - g$
$(13,\ 146/11]$	$73 - 11x/2$	Theorem 2.1(c), second term
$(341/25,\ 69/5]$	$11x/2 - 75$	Section 4.7(v), family Z
$(69/5,\ 14]$	$x/2 - 6$	Section 4.7(v), family Z
$(14,\ 711/50]$	$64 - 9x/2$	Section 4.7(v), family Z
$(30287/2000,\ 78/5]$	$7x/2 - 53$	Section 4.7(v), family Z
$(78/5,\ 16]$	$25 - 3x/2$	Section 4.7(v), family Z
$(16,\ 809/50]$	$73 - 9x/2$	Section 4.7(v), family Z
$(809/50,\ 178/11]$	$73 - 9x/2$	Section 4.8
$(178/11,\ 1619/100]$	$x - 16$	Section 4.8
$(1619/100,\ 118/7]$	$x - 16$	Section 4.7(iii)
$(118/7,\ 69/4]$	$9x/2 - 75$	Theorem 2.1(d), $(9p - 2C)/2$
$(69/4,\ 35/2]$	$x/2 - 6$	Theorem 2.1(d), $(p-1)/2 + h - b$
$(35/2,\ 128/7]$	$64 - 7x/2$	Theorem 2.1(d), $(2(n+q+h) - 7p)/2$
$(64/3,\ 23]$	$3x - 64$	Theorem 2.1(a), $3p - (n+q+h)$
$(23,\ 24]$	5	Theorem 2.1(a) up to $x = 70/3$, then (b)
$(24,\ 26]$	$53 - 2x$	Theorem 2.1(b), $n - g - 2p$
$(26,\ 53/2]$	$53 - 2x$	Section 4.7(i)

The six rows on $(341/25, 809/50]$ come from the family Z of Section 4.7(v), in two forms, on $(341/25, 711/50]$ and on $(30287/2000, 809/50]$. The areas of the rows of the table add up to $21.02610\ldots$, and the 84 intervals below 12 contribute $1.03361\ldots$, so that the total is $22.05972\ldots$. The exact sum is computed in the formalization from the list of intervals.

For Theorem 2.1 one checks the hypotheses on the intervals above. The hypotheses $e \geq 0$ and $n \leq 4p + 2g + 1$ both read $x \geq 12$, and $L \leq 6p$ reads $x \geq 35/3$. So Theorems 4.4 and 4.6 apply for $x \geq 12$. For $12 \leq x \leq 13$ one has $\mu_0 = H + q - 6p = 81N - 6p$, so the first branch of Theorem 4.4 applies, and for $x \geq 13$ one has $\mu_0 = 3N$ and the second branch, which is positive until $x = 146/11$. In Theorem 4.6, $m_1 = p$ for $x \leq 35/2$ and $m_1 = 70N - 3p$ above, and $m_0 = 81N - 4p$ for $x \leq 69/4$ and $12N$ above. The value $m_1 - m_0 - e$ is $9x/2 - 75$, $x/2 - 6$ and $64 - 7x/2$ on the three intervals $(50/3, 69/4]$, $(69/4, 35/2]$, $(35/2, 128/7]$, and it is non-positive outside $(50/3, 128/7)$. Theorem 4.1 applies for $x > 10$ and $x \geq 39/2$, and $D_T = \min(5N, 52N - 1 - 2p, 75N - 3p)$. It gives $3p - (n+q+h) = 3x - 64$ up to $x = 23$ and the cap $g = 5N$ up to $x = 70/3$. From $x = 70/3$ the hypotheses of Theorem 4.3 hold up to $x = 26$ (they read $3p \geq L$, $3p \leq H$, $2p \leq n + q$ and $2h \leq p + 1$), and give $\min(n, 2p + 2g + 1) - g - 2p - 1$, which is $5N$ for $x \leq 24$ and $53N - 2p - 1$ above. In each case the value of the theorem is the value in the table minus at most one unit, from the integer rounding. The same kind of check, with linear inequalities in N, p and the parameters, gives the bounds of Section 4.7 on their intervals, and Section 4.8 gives the others. This proves Theorem 2.4. The loss of 2 in its statement covers the integer rounding of all the bounds.

Remark 4.7. Numerical computations of the exact rank of $\mathcal{A}^0_{N,p}$ for several designs and primes show that the bounds of Theorem 2.1 are attained in the windows where they were tested, and that other families add kernel where the windows overlap. These computations are not part of the proof. An upper bound for the nullity is not proved, and none is needed.

Remark 4.8. The subspaces V of Section 4.8 were found numerically, as optimal constant subspaces for the bound of Theorem 2.3 on each window. They are dense rational subspaces with no evident closed form, which is why they are stored. Only the stored integer identities and the checked window conditions enter the proof.

5. The real place: perturbation and fixed columns

In this section and the next we prove Theorem 2.7. The determinant Δ_N must be small not only at G but on an interval of length $e^{-\rho N}$ around it. Here we reduce this to determinants in which some columns of $\mathcal{A}_N$ are replaced by values of the row functions at fixed points (§5.1–5.2). We then recall OpenAI's energy method, extended to configurations with fewer free points (§5.3). It splits into two cases. In the first case it already gives the decay rate of Theorem 2.7. In the second case it falls short by the gain 8.29824 when few columns are fixed. Section 6 closes that gap. The section ends with the trial sequences and the verification of the bounds (§5.5–5.7).

The design is a parameter throughout. We write $\alpha_A = A/n = (b_0 + q_0 + 2g_0)/n_0$, $\beta_h = 2h_0/n_0$, $\gamma_H = H/n = (n_0 + b_0 + q_0 + g_0 + h_0)/n_0$, $\gamma_d = d_0/n_0 = (n_0 - g_0)/n_0$, and

$$c_{07} = 1 - \frac{b_0 + q_0 + g_0}{n_0}, \qquad c_{26} = 3 - \frac{b_0 + q_0 + 2g_0}{n_0}.$$

For D these are $\alpha_A = 11/29$, $\beta_h = 3/29$, $\gamma_H = 39/29$, $\gamma_d = 53/58$, $c_{07} = 41/58$ and $c_{26} = 76/29$.

5.1. A quadrature formula for the coefficient of the constant.

Let $\delta = x - G$. Then $\Delta_N(x) = \det(\mathcal{A}_N + \delta\mathcal{L}_N)$. The question is how large the columns of $\mathcal{L}_N$ are compared with those of $\mathcal{A}_N$. By (3.3) the entries of $\mathcal{L}_N$ are linear functionals of the coefficients of the row polynomials. These polynomials have bounded degree. So the functionals can be written exactly as finite combinations of point evaluations. If $\tau_1, \dots, \tau_J$ are distinct points and J is at least the number of degrees of freedom of the rows, Lagrange interpolation gives

$$\mathcal{L}_N(r,k) = \sum_{a=1}^{J} A_r(\tau_a)\,\beta_{a,k}, \tag{5.1}$$

where the weight $\beta_{a,k}$ is the value of the same functional on the a-th Lagrange basis polynomial. This is a Gauss-type quadrature formula. It makes the columns of $\mathcal{L}_N$ combinations of point evaluations of the row functions that

define $\mathcal{A}_N$. The perturbed determinant can then be treated by the energy method. The cost is the size of the weights.

We take $J = d_0 N = 53N$ nodes. Put $g_* = 401/2500$ and $a_* = 2g_*/(1 + g_*^2) = 2005000/6410801$, and let $K = \lceil J/2 \rceil$. The nodes are the points

$$t_{j,\pm} = \pm\sqrt{a_*^2 + (1 - a_*^2)(1 + s_j)/2}, \qquad T_K(s_j) = 0,$$

that is, the roots of $T_K\big((2t^2 - (1 + a_*^2))/(1 - a_*^2)\big)$. We keep all K positive roots and the first $J - K$ negative ones, so that for odd J exactly one root is dropped. The nodes lie in $S = (-1, -a_*) \cup (a_*, 1)$. In the coordinate x with $t = 2x/(1 + x^2)$ every node has $|x| > g_*$. This matters below, because the energy fields are larger near $x = 0$. Dropping a root changes the nodal polynomial by a linear factor. It changes the derivative bound and the size of the Lagrange polynomials by factors polynomial in N only. So an odd number of nodes costs nothing in the exponent. This is needed, since $J = 53N$ is odd for odd N.

Let g_S be the Green function of $\mathbb{C} \setminus \bar{S}$ with pole at infinity. In closed form

$$g_S(z) = \tfrac{1}{2} \operatorname{arccosh}\Big(\frac{|z^2 - 1| + |z^2 - a_*^2|}{1 - a_*^2}\Big).$$

The Lagrange polynomials satisfy $|\ell_a(z)| \le e^{53 N g_S(z) + o(N)}$ at distance at least c/N from the nodes. This follows from the derivative identity for Chebyshev polynomials at their roots. The weight of column k is a coefficient functional of ℓ_a multiplied by the column filter. After removal of the common factor of the rows it equals

$$\lambda_k = 4\,[t^0]\,F_k(t)\,(1 - t^{-2})^{-1/2}, \qquad F_k(t) = t^{\,A-b-q-k}(1 - t)^h (t - 1)^q \ell_a(t),$$

where $(1 - t^{-2})^{-1/2} = \sum_l c_l t^{-2l}$ with $c_l = \binom{2l}{l} 4^{-l}$. The next lemma bounds it by a contour that depends on k.

Lemma 5.1 (keyhole bound)**.** *Let P be a real polynomial, $J \ge 0$, and $F(t) = t^{-J} P(t)$. Let $0 < r \le 1$, and suppose that $|F| \le M$ on the circle $|t| = r$ and $|F| \le M'$ on the segments $r \le |x| \le 1$. Then*

$$\Big|4 \sum_{l \ge 0} c_l\,[t^{J+2l}]P\Big| \le 4(1 + J)\max(M, M').$$

Proof. Write $F = \Pi + R$ with Π a polynomial and $R = \sum_{i=1}^{J} \pi_i t^{-i}$ the principal part at 0. The functional sees only Π. Since $c_l = \frac{1}{\pi}\int_{-1}^{1} x^{2l}(1 - x^2)^{-1/2} dx$, its value is $\frac{4}{\pi}\int_{-1}^{1} \Pi(x)(1 - x^2)^{-1/2} dx$, which is at most $4 \max_{[-1,1]} |\Pi|$. By Cauchy's estimate $|\pi_i| \le M r^i$, so $|R(x)| \le JM$ for $r \le |x| \le 1$, and $|\Pi| \le M' + JM$ there. For $|x| < r$ the maximum principle gives $|\Pi(x)| \le \max_{|t|=r} |\Pi| \le (1 + J)M$. □

In the formal proof the integral is replaced by the finite Gauss–Chebyshev identity $\frac{1}{M}\sum_{j<M} \cos^{2l} \frac{(2j+1)\pi}{2M} = c_l$ for $l < M$, and the Cauchy estimate by

the finite Cauchy formula at roots of unity. So no integration is needed. With $\kappa = k/n$, Lemma 5.1 gives

$$|\lambda_k| \le e^{(w(\kappa)+o(1))N}, \qquad w(\kappa) = \max_{z\in\Gamma_r} \big[(10-58\kappa)\log|z| + 6\log|1-z| + 53 g_S(z)\big], \tag{5.2}$$

where Γ_r is the circle $|z| = r$ together with the segments $r \le |x| \le 1$. Here $10 = (A-b-q)/N$ and $6 = (q+h)/N$. The radius r may depend on κ. For κ near 1 the factor $|z|^{10-58\kappa}$ favours a large circle, and for small κ a small one. A single circle of radius 1.001 gives $w(\kappa) \le 52.785$ for every κ. The keyhole contour gains only a little over it at the worst column. Its value is that the worst column for the weight is not the worst column for the binomial factor below.

5.2. **Column expansion and an identity of Andréief type.** Write $\Phi(r,a) = A_r(\tau_a)$. Expanding $\det(\mathcal{A}_N + \delta\Phi\beta)$ multilinearly in the columns gives a sum over maps f from the columns to $\{\varnothing, 1, \dots, J\}$ of $\prod_{f(k)\neq\varnothing} \delta\beta_{f(k),k} \det M_f$. Here M_f is the matrix whose k-th column is the k-th column of $\mathcal{A}_N$ if $f(k) = \varnothing$, and the vector $(A_r(\tau_{f(k)}))_r$ otherwise.

Lemma 5.2. *Suppose that* $|\det M_f| \le M \prod_{f(k)\neq\varnothing} C_k$ *for every* f*, and that* $|\beta_{a,k}| \le B_k$*. Then*

$$\big|\det(\mathcal{A}_N + \delta\Phi\beta) - \det \mathcal{A}_N\big| \le M\Big(\prod_k (1 + J|\delta| B_k C_k) - 1\Big).$$

The lemma allows a different constant for each column. Each M_f has $p = n - m$ columns that are integrals and m columns that are point values. Andréief's identity extends to this situation.

Lemma 5.3. *Let* μ, ν *be finite measures,* F_r $(r < p + m)$*,* G_i $(i < p)$ *and* K *integrable functions, and let* B *be the matrix with* $B(r,i) = \iint F_r(t) K(t,s) G_i(s)\, d\mu(t)\, d\nu(s)$ *for* $i < p$ *and* $B(r, p+j) = F_r(\tau'_j)$*. Then*

$$(p!)^2 \det B = \iint \det\big[F_r((t\|\tau')_c)\big]_{r,c} \cdot \det[K(t_i,s_j)]_{i,j<p} \det[G_i(s_j)]_{j,i<p}\, d\mu^p(t)\, d\nu^p(s),$$

where $t\|\tau'$ *is the list of the* p *variables followed by the* m *fixed points.*

Proof. For $m = 0$ this is Andréief's identity applied twice [12]. In general, expand $\det B$ multilinearly in its first p columns. This introduces one pair of variables (t_i, s_i) for each of them. Then antisymmetrize, first in the t variables and then in the s variables. The last m columns are constant throughout. □

With $K(t,s) = 1/(1-ts)$ the middle determinant is a Cauchy determinant. The third is a generalized Vandermonde determinant $\prod_j s_j^b (1-s_j)^q \det[s_j^{E_i}]$, where the exponents $E_i < n$ are those of the columns that remain integrals.

The missing exponents are the replaced columns. We bound the generalized Vandermonde determinant by a product over the missing exponents.

Lemma 5.4. *Let $s_1, \dots, s_p \in [0,1]$, let $E_1 < \cdots < E_p < n$ be integers, and let $\{k_1, \dots, k_m\}$ be the complement of $\{E_i\}$ in $\{0, \dots, n-1\}$. Then*

$$|\det[s_l^{E_i}]| \le \prod_{j=1}^{m} \beta(k_j) \prod_{i<l} |s_i - s_l|, \qquad \beta(k) = \sum_{i=0}^{m-1} \binom{p}{p-k+i},$$

with $\binom{p}{r} = 0$ for $r \notin [0,p]$.

Proof. Let $P(X) = \prod_l (X - s_l) = \sum_r (-1)^r e_r(s) X^{p-r}$. Multiplying the matrix by a unitriangular matrix built from P gives $|\det[s_l^{E_i}]| = \prod_{i<l} |s_i - s_l| \cdot |\det Q|$, where Q is the $m \times m$ matrix with entries $Q_{j,i} = [X^{k_j - i}]P$. So $|Q_{j,i}| = e_{p-k_j+i}(s) \le \binom{p}{p-k_j+i}$, since $s \in [0,1]^p$. Hadamard's inequality, applied to the rows of Q in the ℓ^1 norm, gives the claim. □

The bound does not depend on the points s_l. Since $\binom{p}{r} \le \binom{n}{r}$, one has $\log \beta(k) \le n_0 \tilde{H}(k/n) N + O(\log N)$, where $\tilde{H}(\kappa)$ is the binary entropy $H(\kappa)$ for $\kappa \ge 1/2$ and $\log 2$ for $\kappa < 1/2$. The weight (5.2) grows with κ up to $\kappa \approx 0.84$, while $\tilde{H}$ is largest at $\kappa = 1/2$. So the product of the two factors is smaller than the product of their maxima.

5.3. **The energy estimate with fixed points.** OpenAI bounds the integrand in Andréief's identity by the exponential of $n^2 E$, where E is a discrete logarithmic energy of the points x_i and s_j in the coordinate $t = 2x/(1+x^2)$. The energy consists of an external field for each point, logarithmic interactions $\log|x_i - x_j|$, $\log(1 - x_i x_j)$ and $\log|s_i - s_j|$ among points of the same kind, and an interaction $-\log(1 - 2x_i s_j + x_i^2)$ between the two kinds. There are two cases, according to which of OpenAI's two bounds for the determinant of the row functions is used [26, Props. 5.2, 5.3]. In the second case the interaction among the x_i has weight 2 instead of 1. In the first case the row field acquires a term with a multiplier λ_1 for the constraint that defines the case. The second case also bounds a determinant $\det \mathsf{L}$ of size n by 6^n. This is the factor that Section 6 improves.

The energy is bounded from above by duality. Given trial sequences $u = (u_k)$ and $v = (v_k)$, one writes the interaction terms as quadratic forms in the moments $\frac{1}{n}\sum_i T_k(x_i)$ and $\frac{1}{n}\sum_j s_j^k$. Then one completes the squares against u and v. What remains is the sum of the norms $\|u\|^2 = \sum_k u_k^2/k$ and $\|v\|^2$ and, for each point, a function of one variable. With $\mathsf{T}(u,x) = \sum_k u_k T_k(x)/k$ and $\mathsf{S}(v,x) = \sum_k v_k x^k/k$ these are, in the second case,

$$X(x) = \alpha_A \log|x| + \beta_h \log(1-x) - \gamma_H \log(1+x^2) - 4\mathsf{T}(u,x) - \mathsf{S}(v,x),$$
$$Y(s) = \tfrac{b_0}{n_0} \log s + \tfrac{q_0}{n_0} \log(1-s) + 2\mathsf{T}(v,s).$$

In the first case they are

$$X_1(x) = \alpha_A \log|x| + \beta_h \log(1-x) - (\gamma_H - \tfrac{1}{2})\log(1+x^2) \\ + \lambda_1(\tfrac{2g_0}{n_0} - \tfrac{2x^2}{1+x^2}) - 2\mathsf{T}(p,x) - \mathsf{S}(v,x),$$

with the same Y. The trial sequence for the first variables is called p in this case. If $X \le M_X$ and $Y \le M_Y$ on the whole domains, the energy of every configuration is at most a constant determined by M_X, M_Y and the norms. In potential-theoretic terms the extremal configuration is described by an equilibrium measure for two interacting measures in an external field [32]. The best trial sequences are its moments, as Carreira observed [7]. Any trial sequences give a valid upper bound.

When m of the n columns are point values, the configuration has only $n-m$ free points of each kind. The normalization is still by n. The fixed points enter through the determinant of the row functions evaluated at the list $t\|\tau'$. They are of two kinds. Some are quadrature nodes, with $|x| > g_*$. The others are points $y \in (0,1)$. We write $\alpha = m/n$ for the fraction of fixed columns and τ for the fraction of fixed points of the second kind. The missing free points change the energy bound by an amount linear in m. This is the energy cost of a fixed column. Every step of OpenAI's argument remains valid uniformly in the number of free points. The regularization of the logarithmic interactions, the diagonal corrections and the completion of squares only involve sums over the points present. For the fixed points of the first kind the relevant field is the residual field

$$R(x) = -\gamma_d \log(1+x^2) - 2\mathsf{T}(u,x), \qquad g_* \le |x| \le 1.$$

For those of the second kind it is

$$N(y) = -\gamma_d \log(1+y^2) + \frac{n_0 - 2g_0 - 1}{n_0}\log y - u_1 y, \qquad 0 < y < 1,$$

where u_1 is the first coefficient of u. The coefficient $(n_0 - 2g_0 - 1)/n_0$ is a lower bound, valid for all $N \ge 1$, of $(n-1-2g)/n$.

With Lemma 5.4 the generalized Vandermonde determinant contributes only the factor $\prod_{i<l}|s_i - s_l|$ to the energy. The binomial factors are charged to the columns. The energy bound then reduces to a quadratic inequality in α and τ.

Lemma 5.5 (budget, second case)**.** *Let $\ell = \log 2$, and let $\mathcal{N}, \Phi, \mathcal{R}, \mathcal{N}_c$, c_2 and $m \ge 0$ be real numbers with*

$$\mathcal{N} + \Phi + U - c_{07}\ell \le -m, \qquad \ell - \mathcal{R} + \mathcal{N}_c \le 0,$$
$$\big((c_{26}-1)\ell - \Phi + \mathcal{R} - c_2\big)^2 \le 4\ell m.$$

Then for all $\alpha, \tau \ge 0$

$$(-(c_{07} - \tfrac{1}{2}) + (c_{26}-1)\alpha - \alpha^2 + \tau - \tau^2)\ell + \mathcal{N} + (1-\alpha)\Phi \\ + (\alpha - \tau)\mathcal{R} + \tau\mathcal{N}_c \le -U + \tfrac{1}{2}\ell + c_2\alpha.$$

Proof. The left side minus the right side equals $(\mathcal{N}+\Phi+U-c_{07}\ell)+\alpha\big((c_{26}-1)\ell-\Phi+\mathcal{R}-c_2\big)-\ell\alpha^2+\tau(\ell-\mathcal{R}+\mathcal{N}_c)-\ell\tau^2$. The last two terms are non-positive by the third hypothesis. The first three form a concave quadratic in α. It is non-positive for all α, because $a+s\alpha-\ell\alpha^2\le 0$ for all α as soon as $s^2+4\ell a\le 0$. □

The lemma is applied with $\mathcal{N}$ a bound for $2\|u\|^2+\frac12\|v\|^2$, $\Phi=M_X+M_Y$, $\mathcal{R}$ a bound for the supremum of R and $\mathcal{N}_c$ a bound for the supremum of N. The analogue for the first case has no fixed points of the second kind. Its polynomial in α is

$$(-(c_{07}-\tfrac12)+(c_{26}-1)\alpha-\alpha^2)\ell+n_1+(1-\alpha)F_1+\alpha\lambda_1\tfrac{2g_0}{n_0}\le -U+\tfrac12\ell+c_1\alpha,$$

with n_1 a bound for the norms of the first trial and $F_1=M_{X_1}+M_Y$. It holds for all real α when its discriminant is non-positive and $c_1\le(c_{26}-1)\ell-F_1+\lambda_1\frac{2g_0}{n_0}$.

5.4. **The two cases at the improved rate.** Theorem 2.7 asks for the rate $U_0=U+8.29824/n_0^2$ instead of U. A larger target is a smaller slack m in Lemma 5.5. In the first case the trial of §5.5 has enough slack: its budget holds at U_0 for every α, with the cost c_2 of the second case. In the second case the budget at U_0 fails for small α and holds for $\alpha\ge 1/16$. The reason is that the constant U_0-U is absorbed by the linear term once α is not small. So only the second case with fewer than $n/16$ fixed columns needs a new argument. That argument is Section 6. It proves the bound of the second case at U_0 with the same cost c_2, for $0\le\alpha\le 1/16$.

Assume this for the moment. A determinant M_f with m fixed columns $k_1,\dots,k_m$ is then at most $C^m\prod_j\beta(k_j)e^{n_0c_2Nm}$ times $\exp\big(n_0^2N^2(-U_0+\frac12\log 2+\varepsilon)\big)$, with an absolute constant C. The factor C^m comes from the expansion of the row functions at the fixed points. By (5.2) and the bound for $\beta(k)$, the constants of Lemma 5.2 satisfy

$$\log(B_kC_k)\le\big(n_0c_2+w(k/n)+n_0\tilde H(k/n)+o(1)\big)N\le(\rho+o(1))N,$$

with $\rho=n_0c_2+\Psi_0$ and $\Psi_0\ge\sup_\kappa\big(w(\kappa)+n_0\tilde H(\kappa)\big)$. For $|\delta|\le e^{-\rho N}$ Lemma 5.2 gives a factor $(1+O(N^5))^n=e^{o(N^2)}$. This proves Theorem 2.7, given Section 6 and the numerical bounds below.

5.5. **The trial sequences.** For the first case we use a trial sequence with 40 Chebyshev coefficients and a finite number of geometric terms, with the multiplier $\lambda_1=1.68493357$. For the second case we use trial sequences of the form

$$u_k=c_k+\operatorname{Re}\sum_j r_jz_j^k\qquad(k\ge1;\ c_k=0\text{ for }k>K),\tag{5.3}$$

with $K=160$ Chebyshev coefficients, which are dyadic rationals, and 75 geometric terms. Of these, 41 have complex z_j and complex r_j.

The geometric terms have a simple meaning. Since $\sum_{k\ge1} z^k T_k(x)/k = -\frac{1}{2}\log(1-2xz+z^2)$, a term (z,r) contributes $-\frac{1}{2}\operatorname{Re} r\log(1-2xz+z^2)$ to the potential $\mathsf{T}(u,x)$. If $x_0+i\epsilon = \frac{1}{2}(z+z^{-1})$, then $1-2xz+z^2 = 2z(x_0+i\epsilon-x)$. So the term is, up to a constant, the logarithmic potential of a point charge of complex strength r at $x_0+i\epsilon$ and its conjugate. The extremal measure has a gap in its support around $x=0$. Its potential has a logarithmic singularity at $x=0$ and square-root behaviour at the edges of the gap. A Chebyshev expansion converges slowly near such points. A charge at a small distance ϵ reproduces the local behaviour with one term. We place the complex charges at the 41 equally spaced points $x_0 = -0.16 + 0.008j$ $(0\le j\le 40)$, all with $\epsilon = 10^{-2}$. The real terms sit at points $\pm z$ with z increasing from 0.8 to 0.999997. The numbers z_j are rationals with at most twelve decimals. The norms of the geometric terms have closed forms. Two terms (z_j,r_j) and (z_l,r_l) contribute $-\frac{1}{2}\operatorname{Re}(r_j r_l\log(1-z_jz_l) + r_j\bar r_l\log(1-z_j\bar z_l))$ to $\|u\|^2$.

5.6. **Verification of the bounds.** We must bound three functions of one variable from above on their whole domains. They are the field X on $[-1,1]$, the field Y on $[0,1]$, and the residual field R on $g_*\le|x|\le1$. With $x=\cos\theta$, each is a constant plus a cosine series with K terms plus finitely many terms $\operatorname{Re} w\log(1-ae^{im\theta})$.

We cover each domain by cells $[x_c-h, x_c+h]$ whose endpoints are multiples of 2^{-40}. On each cell we bound the function by its Taylor expansion at x_c. The cells are grouped in blocks. For each block the derivatives of the polynomial part are computed exactly at the centre of the block up to order 70. The remainder beyond order 70 is bounded by a single constant for the block, multiplied by $h^{70}/70!$. The Taylor expansion at the centre of a cell is obtained from that of the block by an exact shift. The logarithmic terms are expanded to order 5, with remainders bounded from the exact minimum of $|1-ae^{im\theta}|$ on the cell. A few cells at $x=0$ and $x=\pm1$ treat the singular terms separately. The covers of X, Y and R have 47, 38 and 23 blocks and 373, 302 and 176 cells. Each cell is checked by exact rational and integer arithmetic.

The resulting bounds are

$$\sup X \le -0.96450499, \quad \sup Y \le -1.57499400, \quad \sup_{g_*\le|x|\le1} R \le -0.0631062.$$

So $M_X + M_Y = -2.53949899$, and the norms satisfy $2\|u\|^2 + \frac{1}{2}\|v\|^2 \le 0.7736205195$. The first coefficient satisfies $u_1 \ge 9/50$, which is used near $x=0$. For the points of the second kind a one-variable estimate shows $\sup_{0<y<1} N(y) \le -77/100$ for $u_1 \ge 9/50$. For the first case, with the weights $(\alpha_A, \beta_h, \gamma_H - \frac{1}{2}, \lambda_1, 2g_0/n_0) = (11/29, 3/29, 49/58, \lambda_1, 5/29)$, the bounds $\sup X_1 \le -1.15854$ and $\sup Y \le -1.32981$ are obtained by isolating the zeros of the numerators of the derivatives. These numerators are polynomials with integer coefficients. The norm bound is $n_1 = 0.67761$.

With $c_{07} = 41/58$, $c_{26} = 76/29$ and $U = 2.255858836$, the hypotheses of Lemma 5.5 hold with $m = 2.986 \cdot 10^{-6}$ and $c_2 = 3.596893$. The first case admits $c_1 = 3.549542 < c_2$. At the target U_0 the first-case budget still holds with the cost c_2, and the second-case budget holds for $\alpha \geq 1/16$ with the cost c_2. The trial data are fixed once and for all. The file containing them has SHA-256 digest

`ff307ca0287dc5224e268202cb0b09776e933652acd2f2f879492af28d3e381d.`

5.7. **The perturbation exponent.** It remains to bound Ψ_0. For a fixed radius the function $\kappa \mapsto \max_{\Gamma_r} \Phi_\kappa$ is a maximum of functions affine in κ, hence convex. The entropy H is concave. On a cell $[\kappa_1, \kappa_2] \subset [\frac{1}{2}, 1]$ we replace H by its tangent at the midpoint. The sum is then convex, and its supremum on the cell is attained at an endpoint. On $[0, \frac{1}{2}]$ the term $\tilde{H} = \log 2$ is constant. We cover $[0, 1]$ by 14 cells, each with a rational radius r. At each endpoint the maximum of Φ_κ on the circle and on the segments is bounded by a cover of the contour by small arcs and intervals, in exact arithmetic. This proves

$$\sup_{0 \leq \kappa \leq 1} \big(w(\kappa) + 58\tilde{H}(\kappa)\big) \leq \Psi_0 = 87.9663.$$

The supremum is attained near $\kappa = 0.62$, with radius about 0.6. Hence $\rho = 58 \cdot 3.596893 + 87.9663 = 296.586094$.

6. The real place: a signed bound for the second case

In this section we prove the bound of the second case at the rate U_0, for configurations with at most $n/16$ fixed columns. This completes the proof of Theorem 2.7. The section is organized as follows. The second case contains a determinant $\det \mathsf{L}$ of size n that the energy method bounds by 6^n. We first write it with Pick matrices (§6.1). A Schur complement argument and a contour bound reduce it to the block of the negative nodes (§6.2–6.3). Heine's formula turns that block into a ratio of two integrals of log-gas type (§6.4). We bound the numerator from above and the denominator from below (§6.5). Positive nodes close to 1 need a separate argument (§6.6). Combined with OpenAI's energy, the bounds give a concave majorant thanks to an inequality for signed logarithmic energy (§6.7). Fixed columns add fixed rows (§6.8). The hypotheses of the reduction hold on all configurations that matter (§6.9). The section ends with the trials and their verification (§6.10).

6.1. **The determinant of the row functions.** In the second case OpenAI's integrand contains the determinant of the row functions at the points $x_1, \dots, x_n \in (-1, 1) \setminus \{0\}$ [26, Prop. 5.3]. With $e = n - 1 - 2g$ it factors as

$$\det \varphi(x) = c\, V(x) \prod_a |x_a|^{-(n-1-g)} \det \mathsf{L}(x), \tag{6.1}$$

where c depends only on n and g, and $V(x) = \prod_{a<b}(x_b - x_a)$. To describe L let

$$B(z) = \prod_a \frac{z - x_a}{1 - x_a z}, \qquad \mathcal{K}_{ab} = \frac{1}{1 - x_a x_b},$$

be the Blaschke product of the nodes and the Pick (Cauchy) matrix of the Szegő kernels $k_a(z) = 1/(1 - x_a z)$. Put $w_a = 1$ at a negative node and $w_a = -5$ at a positive node, and $D = \operatorname{diag}(w_a B'(x_a) x_a^e)$. Then $\det \mathsf{L} = \det(\mathcal{K} + D)/\det \mathcal{K}$. In the basis k_a the operator L acts on the model space $\mathcal{K}_B = H^2 \ominus BH^2$, which has dimension n. The energy method uses only $\|\mathsf{L}\| \le 6$, which gives $|\det \mathsf{L}| \le 6^n$. On the configurations near the extremal one, the true size of $\det \mathsf{L}$ is smaller by a factor e^{-cN^2}. We now capture part of this.

Let k and q be the numbers of negative and positive nodes. For a weight $\omega \in \mathbb{R}$ let $F(\omega) = \mathcal{K} + \operatorname{diag}(D_-, \omega D_+)$, where D_- and D_+ are the diagonal blocks of $\operatorname{diag}(B'(x_a)x_a^e)$ at the negative and the positive nodes. So $\det \mathsf{L} = \det F(-5)/\det \mathcal{K}$. The case $\omega = 1$ is a Pick matrix.

Lemma 6.1. *For every configuration $0 \le F(1) \le 2\mathcal{K}$.*

Proof. For a real vector c put $\phi(z) = B(z) \sum_a c_a/(z - x_a)$. This is the function of $\mathcal{K}_B$ with node values $\phi(x_a) = c_a B'(x_a)$, and $\|\phi\|_{H^2}^2 = c^T \mathcal{K} c$. By the residue theorem on the unit disc

$$c^T (F(1) - \mathcal{K}) c = \sum_a \frac{x_a^e \phi(x_a)^2}{B'(x_a)} = \frac{1}{2\pi i} \oint_{|z|=1} \frac{z^e \phi(z)^2}{B(z)}\, dz.$$

On the unit circle $|z^e \phi^2 / B| = |\phi|^2$, so the right side lies in $[-c^T\mathcal{K}c,\, c^T\mathcal{K}c]$. □

6.2. **A Schur complement bound.** The determinant $\det F(-5) = \det(F(1) - 6\operatorname{diag}(0, D_+))$ is a perturbation of a Pick matrix in the positive block only. The following inequality bounds it by the negative block.

Theorem 6.2. *Let $M = \begin{pmatrix} A & C^* \\ C & E \end{pmatrix}$ be a Hermitian block matrix with $0 \le M \le b$ and $A > 0$. Let T be Hermitian of the size of E, with $-\tau \le T \le \tau$, and let $c \ge 0$. Then*

$$\left|\det \begin{pmatrix} A & C^* \\ C & E - cT \end{pmatrix}\right| \le (b + c\tau)^{\dim E} \det A.$$

Proof. The determinant equals $\det A \cdot \det(S - cT)$, where $S = E - CA^{-1}C^*$ is the Schur complement. Since $M \ge 0$ one has $S \ge 0$, and $S \le E \le b$. So $-c\tau \le S - cT \le b + c\tau$, and every eigenvalue of $S - cT$ has modulus at most $b + c\tau$. □

We apply this in the metric of $\mathcal{K}$, that is, after the congruence by a Cholesky factor of $\mathcal{K}$. By Lemma 6.1 the hypotheses hold with $b = 2$ and

$c = 6$, provided that $\pm D_+ \le \tau\mathcal{K}$ as quadratic forms. The result is

$$|\det \mathsf{L}| \le (2+6\tau)^q \det \mathsf{L}_{--}, \qquad \det \mathsf{L}_{--} = \frac{\det(\mathcal{K}_{--} + D_-)}{\det \mathcal{K}_{--}}, \tag{6.2}$$

where $\mathcal{K}_{--}$ is the block of $\mathcal{K}$ at the negative nodes. The factor $(2+6\tau)^q$ is $e^{O(N)}$. All the information of order N^2 is in $\det \mathsf{L}_{--}$, which depends on the positive nodes only through the values $B_+(x_b)$ of their Blaschke product at the negative nodes.

6.3. **A contour bound.** It remains to bound τ. The quadratic form of D_+ is the sum of the residues of $z^e\phi^2/B$ at the positive nodes. These lie in the right half of the disc. We integrate over the boundary of the right half-disc instead of the circle.

Lemma 6.3. *Let $M_* = \sup_{-1<y<1} |y|^e/|B(iy)|$. For every real vector c*

$$\Big|\sum_{x_a>0} B'(x_a) x_a^e c_a^2\Big| \le \Big(1 + \frac{M_*}{2}\Big) c^T \mathcal{K} c.$$

If $\frac{e}{n} - \frac{1}{n}\sum_a \Big(\frac{t^2}{t^2+x_a^2} - \frac{x_a^2 t^2}{1+x_a^2 t^2}\Big) > 0$ for every $t \in (0,1]$, then $M_* = 1$, and the bound is $\frac{3}{2} c^T \mathcal{K} c$.

Proof. With ϕ as in the proof of Lemma 6.1, the sum is the contour integral of $z^e\phi^2/B$ over the boundary of the right half-disc divided by $2\pi i$. No node lies on this boundary. On the arc the integrand has modulus $|\phi|^2$, and the arc contributes at most $\|\phi\|^2$. On the diameter $z = iy$ the modulus is at most $M_*|\phi(iy)|^2$. By the Fejér–Riesz inequality on a diameter, $\int_{-1}^{1} |\phi(iy)|^2 dy \le \pi\|\phi\|^2$. So the diameter contributes at most $\frac{1}{2} M_* \|\phi\|^2$.

For the second statement write $\log(|y|^e/|B(iy)|) = n h(y)$. Then $h(y) = \frac{e}{n}\log|y| + \frac{1}{2n}\sum_a \log\frac{1+x_a^2 y^2}{y^2+x_a^2}$, and $h(1) = 0$ because $|B(i)| = 1$. Differentiation gives $t h'(t) = \frac{e}{n} - \frac{1}{n}\sum_a (\frac{t^2}{t^2+x_a^2} - \frac{x_a^2 t^2}{1+x_a^2 t^2})$. If this is positive on $(0,1]$, then $h < 0$ on $(0,1)$. Since h is even, $M_* = 1$. □

The formal proof avoids the residue theorem. It subtracts the poles with divided differences, integrates the remaining holomorphic function over the half-disc with primitives, and proves the Fejér–Riesz step by the same device on the upper half-disc. With $\tau = \frac{3}{2}$, (6.2) becomes

$$|\det \mathsf{L}| \le 11^q \det \mathsf{L}_{--}. \tag{6.3}$$

The condition on t asks that the node measure is not too concentrated near 0 at any scale. Its left side at the extremal measure is at least 0.33 for every t.

6.4. **Heine's formula for the negative block.** For $s \geq 1$ and a positive continuous function f on $[-1,1]$ put

$$Z_s(f) = \int_{[0,\pi]^s} \prod_{l<l'} (\cos\theta_l - \cos\theta_{l'})^2 \prod_l f(\cos\theta_l)\, d\theta.$$

By Heine's formula $Z_s(f)$ is $s!$ times the Hankel determinant of the moments of $f(u)(1-u^2)^{-1/2}du$. Let

$$\phi(u) = \prod_{x_b<0} (1 + x_b^2 + 2x_b u), \qquad \psi(u) = \prod_{x_p>0} (1 + x_p^2 + 2x_p u)^{-1}.$$

Theorem 6.4. *Let $s = q - g$. If $s \geq 1$ and $k \leq e+1$, then*

$$\det \mathsf{L}_{--} = \prod_{b<c}(1 - x_b x_c) \prod_{b,p}(1 - x_b x_p)^{-1} \cdot \frac{Z_s(\phi\psi)}{Z_s(\psi)},$$

where b, c run over the negative nodes and p over the positive nodes.

Sketch. One first writes $\det \mathsf{L}_{--} = \det[x_b^j + B_+(x_b) x_b^{m-j}]_{b,j<k} / V(x_-)$ with $m = e + k - 1$, by Lagrange interpolation at the negative nodes. Since $1 + x^2 + 2xu = x\big(x + x^{-1} + 2u\big)$, the factor ϕ is a polynomial in u with roots $-\frac{1}{2}(x_b + x_b^{-1})$. By Christoffel's formula, $Z_s(\phi\psi)/Z_s(\psi)$ is a determinant of the monic orthogonal polynomials of the weight $\psi(u)(1-u^2)^{-1/2}$ at these roots, divided by a Vandermonde determinant. That weight is of Bernstein–Szegő type: ψ is the reciprocal of $|h(e^{i\theta})|^2$ for the polynomial $h(z) = \prod_p (1 + x_p z)$. Its orthogonal polynomials of degree $\geq q$ are explicit. At $u = -\frac{1}{2}(x + x^{-1})$ they equal Laurent expressions in x built from h. Since $h(z)/z^q \overline{h}(1/z)$ evaluated at $-x_b$ is $B_+(x_b)$ up to sign, the two determinants agree after the Cauchy-type products are collected. □

The formal proof works with the angle variables throughout. It needs Heine's formula, Andréief's identity and the Bernstein–Szegő polynomials, but no integration over orthogonal groups. Both integrals in Theorem 6.4 are of log-gas type. There are s particles $u_l = \cos\theta_l$ on $[-1,1]$ with logarithmic repulsion, in the external field $\log\phi + \log\psi$ or $\log\psi$. Each field is a sum of logarithmic potentials $l(x,u) = \log(1 + x^2 + 2xu)$ of the nodes.

6.5. **Bounds for the two integrals.** The numerator is bounded from above by the method of OpenAI's energy estimate, now with one population. The key identity is $\log|u - v| = -\log 2 - \sum_{k\geq 1} \frac{2}{k} T_k(u) T_k(v)$ for $u \neq v$ in $[-1,1]$. With the moments $M_k = \sum_l T_k(u_l)$ the pair sum is $-\binom{s}{2}\log 2 - \sum_k \frac{1}{k}(M_k^2 - \sum_l T_k(u_l)^2)$. For any sequence a and any $\nu > 0$ one has $-M_k^2 \leq -2\nu a_k M_k + \nu^2 a_k^2$. This completes the square against a.

Lemma 6.5. *Let $\nu \geq s$ and let a be a summable sequence. With $\mathsf{T}(a,u) = \sum_k a_k T_k(u)/k$ and $\|a\|^2 = \sum_k a_k^2/k$,*

$$\log Z_s(f) \leq -s^2 \log 2 + 2\nu^2 \|a\|^2 + s \sup_{[-1,1]} \big(\log f - 4\nu \mathsf{T}(a,\cdot)\big) + o(\nu^2).$$

The diagonal terms $\sum_l T_k(u_l)^2/k$ diverge when summed over k. They are regularized by a damping τ^k with $\tau = 1 - e_N$ as in [26, Prop. 6.1], at a cost $o(\nu^2)$. The completed squares $(M_k - \nu a_k)^2/k$ are dropped in this lemma. They are kept in §6.9.

The denominator is bounded from below by Jensen's inequality. Let ν' be a probability measure on $[0,\pi]$ of the form $\nu' = (1 + 2\sum_{k\le K} b_k \cos k\theta)\,\frac{d\theta}{\pi}$, with a positive density. In the variable $u = \cos\theta$ it is a polynomial multiple of the arcsine law.

Lemma 6.6. *For every bounded measurable Q,*

$$\log Z_s(e^Q) \ge s(s-1)\Big(-\log 2 - \sum_k \frac{2b_k^2}{k}\Big) + s\int Q(\cos\theta)\,d\nu'(\theta) + s\,h(\nu'),$$

where $h(\nu')$ is the entropy of the density of ν' with respect to $d\theta$.

Proof. Apply Jensen's inequality to the product density $\prod_l \frac{d\nu'}{d\theta}(\theta_l)$. The logarithmic energy of ν' in the variable u is $-\log 2 - \sum_k 2b_k^2/k$, by the expansion of $\log|u-v|$ above. □

The fields of the nodes integrate against ν' in closed form. For $|x| < 1$ one has $\int l(x,\cos\theta)\,d\nu' = -2\sum_k (-x)^k b_k/k$, a polynomial in x. The arcsine part contributes 0, because the logarithmic potential of the arcsine law is constant on $[-1,1]$.

6.6. **Positive nodes near 1.** A positive node $x_p = 1-\epsilon$ puts into ψ a factor $(1 + x_p^2 + 2x_p u)^{-1}$ of size ϵ^{-2} near $u = -1$. Lemma 6.5 charges the supremum of the field to all s particles. A single such node would therefore cost about $N\log(1/\epsilon)$, while the true change of $\det \mathsf{L}_{--}$ is bounded. The following reduction removes such nodes from the log gas.

Theorem 6.7. *Let ϕ be positive and non-increasing on $[-1,1]$, and let $\psi = \psi_r \prod_{p\in P} k_p$ with $\psi_r > 0$, $k_p(u) = (1 + 2y_p u + y_p^2)^{-1}$ and $y_p \in (0,1)$. If $|P| = 2m$ or $|P| = 2m-1$, then*

$$\log\frac{Z_s(\phi\psi)}{Z_s(\psi)} \le m\log\phi(-1) + \log\frac{Z_{s-m}(\phi\psi_r)}{Z_{s-m}(\psi_r)}.$$

Sketch. For an odd number of factors add one more factor, at a point to be chosen. Write both integrals as Hankel determinants. Moving all y_p to a common value changes the ratio monotonically. This uses a one-point comparison for the weight $(1 + 2yu + y^2)^{-1}$, whose Chebyshev moments form the geometric sequence $(-y)^k$. For a common value y the $2m$ factors behave like m particles pinned near $u = -1$. Fischer's inequality bounds the numerator by $\phi(-1)^m$ times the determinant of the remaining block. As $y \to 1$ the Schur complement of the pinned block in the denominator converges to the Hankel matrix of the remaining $s-m$ particles. □

We apply the theorem to the positive nodes with $1 - x_p < \theta_0 = 10^{-3}$. Since $\phi(-1) = \prod_b (1-x_b)^2 \le 4^k$, each such pair costs at most $O(N)$. The

energy of OpenAI's estimate charges each such node $\beta_h n \log(1/\theta_0)$, which is larger. So the number of such nodes on the configurations that matter is bounded by a constant times N, with a cost absorbed in the margin.

6.7. **Concavity: an inequality for signed logarithmic energy.** We now combine OpenAI's energy with (6.3), Theorem 6.4 and Lemmas 6.5 and 6.6. The total is a functional of the node measure μ, split into its negative part μ_- and positive part μ_+, and of the column measure. Its quadratic part is not obviously negative. Theorem 6.4 contributes the Cauchy-type term $-\iint \log(1-xy)\,d\mu_-(x)d\mu_+(y)$, and Lemma 6.5 applied with a trial that depends on μ contributes a convex term. Write

$$I(\delta) = -\iint \log|x-y|\,d\delta(x)d\delta(y), \qquad L(\tau) = -\iint \log(1-xy)\,d\tau(x)d\tau(y)$$

for real measures on $[-1,1]$ and $[0,1]$. Both are sums of squares of moments: $L(\tau) = \sum_k \frac{1}{k}\big(\int x^k d\tau\big)^2$. The quadratic part of the combined functional is $-\frac{1}{2}I - \frac{1}{2}\sum_k (B_k - 2C_k)^2/k + \frac{1}{2}L(\mu_+)$, where B_k and C_k are the power moments of the nodes and the columns. The term $\frac{1}{2}L(\mu_+)$ is absorbed by the following theorem.

Theorem 6.8. *Let $\delta = \delta_- + \delta_+$ be a real measure of finite logarithmic energy and total mass 0, with δ_- on $[-1,0]$ and δ_+ on $[0,1]$. Let R be the reflection $x \mapsto -x$. Then*

$$I(\delta) \geq L(\delta_+) + L(R\delta_-) \geq L(\delta_+).$$

Proof. Put $p = \delta_+$ and $r = R\delta_-$, both on $[0,1]$, and let $K_1(a,b) = -\log|a-b|$ and $K_2(a,b) = -\log(a+b)$. Since $|x-y| = a+b$ for nodes of opposite signs, $I(\delta) = K_1(p) + K_1(r) + 2K_2(p,r)$, which equals $\frac{1}{2}(K_1+K_2)(p+r) + \frac{1}{2}(K_1 - K_2)(p-r)$.

First, $p+r$ has mass 0. One has $K_1 = G_{\mathbb{D}} + L$, where $G_{\mathbb{D}}(a,b) = \log|(1-ab)/(a-b)|$ is the Green function of the unit disc, whose energy is non-negative. By Frullani's integral $-\log(a+b) = \int_0^\infty (e^{-(a+b)t} - e^{-t})\frac{dt}{t}$, so $K_2(\sigma) = \int_0^\infty \big(\int e^{-at} d\sigma(a)\big)^2 \frac{dt}{t} \geq 0$ for σ of mass 0. Hence $(K_1+K_2)(p+r) \geq L(p+r)$.

Second, $H = K_1 - K_2 = \log\frac{a+b}{|a-b|}$ satisfies $H(\sigma) \geq L(\sigma)$ for every real σ on $[0,1]$. With $a = e^{-x}$ and $b = e^{-y}$ one has $H = \log\coth\frac{|x-y|}{2} = \sum_{j\ \mathrm{odd}} \frac{2}{j} e^{-j|x-y|}$. For $x, y \geq 0$, $e^{-j|x-y|} = e^{-j(x+y)} + 2j\int_0^{\min(x,y)} e^{-j(x+y-2v)} dv$. Put $F_j(v) = \int_{x \geq v} e^{-j(x-v)} d\sigma$. Then $H(\sigma) = \sum_{j\ \mathrm{odd}} \big(\frac{2}{j} F_j(0)^2 + 4\int_0^\infty F_j(v)^2 dv\big)$. In $L(\sigma) = \sum_k \frac{1}{k}\big(\int e^{-kx} d\sigma\big)^2$ the terms with odd k are $F_k(0)^2/k$. For even $k = j+1$, integration by parts gives $\int e^{-(j+1)x} d\sigma = F_j(0) - \int_0^\infty e^{-(j+1)v} F_j(v)\,dv$. The Cauchy–Schwarz inequality with weights $1+\frac{1}{j}$ and $1+j$ gives $\big(\int e^{-(j+1)x} d\sigma\big)^2 \leq (1+\frac{1}{j})F_j(0)^2 + \frac{1}{2}\int F_j^2$. Summing, $L(\sigma) \leq H(\sigma)$.

Together, $I(\delta) \geq \frac{1}{2}L(p+r) + \frac{1}{2}L(p-r) = L(p) + L(r)$. □

By Theorem 6.8 the quadratic part is at most $-\frac{1}{4}I-\frac{1}{2}\sum_k(B_k-2C_k)^2/k\le 0$ on differences of admissible measures. So the combined functional is concave. Its maximum is bounded by the value at a trial plus the supremum of a linear functional, that is, by a constant plus the suprema of fields of one variable. The formal proof uses a discrete form of Theorem 6.8 for weighted point configurations against a reference measure with bounded density. The fixed rows of the next subsection change the masses of the two species. The missing mass is compensated by phantom particles distributed by the arcsine law, whose potential is constant on $[-1,1]$.

6.8. **Fixed rows.** With m fixed columns the determinant of the row functions is evaluated at the free points and at the fixed points. The fixed points enter $\det \mathsf{L}$ as nodes, but they are not integrated over. Lemma 6.3 then needs a change. Its constant $\frac{3}{2}$ is false when the sum runs over the free positive nodes only, while the Blaschke product and the Gram form contain the fixed nodes as well. For two adjacent positive nodes 0.5 and 0.55 with $e=2$ the best constant is 3.625. The correct statement keeps the fixed positive residues as a separate term $W=\sum x_a^e/((1-x_a^2)|B'(x_a)|)$ over the fixed positive nodes. Each of them is bounded by the reproducing estimate $(c_aB'(x_a))^2(1-x_a^2)\le c^T\mathcal{K}c$.

Instead of absorbing W into τ, we expand. Write $\det(X+\operatorname{diag}\delta)$ as a sum of principal minors over subsets S of the fixed positive nodes. This gives

$$|\det \mathsf{L}|\le 11^q\sum_S\Big(\frac{6}{11}\Big)^{|S|}\det \mathsf{L}_{--}^{(S)}\,W_S,$$

where in the term for S the nodes of S interact with the other nodes through $\log(1-xy)$ instead of $\log|x-y|$. The energy then has three fixed populations: fixed negative nodes, fixed positive nodes, and the nodes of S. Their fields are bounded on their domains like the residual field of Section 5. The fixed nodes are images of quadrature nodes. So the mass of their measure κ near 1 is limited by the spacing of the quadrature nodes, namely $\kappa([x,1])\le C_1(1-x)$ for $x\ge x_c=0.999$. For such a measure of mass α and a field ψ with $\psi\le\lambda$ on $[0,x_c]$ and $\psi(x)\le g_0+r\log\frac{1}{1-x^2}$ on $[x_c,1)$,

$$\int\psi\,d\kappa\le\alpha\lambda+C_1\int_0^{1-x_c}\big(g_0+r\log(1/u)-\lambda\big)_+\,du.$$

The integral has a closed form. This bounds the fixed rows without a finite optimization over their positions.

6.9. **The configurations that matter.** The reduction above needs hypotheses on the configuration. Lemma 6.3 needs the condition on t. The number of positive nodes within θ_0 of 1 must be bounded by a constant times N. The fraction of negative nodes must lie in a fixed window. These hypotheses hold on every configuration whose energy, in OpenAI's estimate, is within the gain of the maximum. We use the following dichotomy. Either OpenAI's bound at the rate U_0 holds for the configuration, and nothing is

to be proved. Or the configuration exceeds it. Then its energy is within $U_0 - U$ of the maximum of OpenAI's bound.

For such a configuration, keep the completed squares and the non-negative differences between the fields and their suprema in the proof of OpenAI's estimate. Their sum is at most the deficit, a constant times $N^2(U_0 - U)$. This bounds the fraction of nodes on which a field is far below its supremum, by Chebyshev's inequality. It also bounds every linear statistic $\sum_k F_k(M_k - \nu a_k)/k$ of the moments, by the Cauchy–Schwarz inequality. The condition of Lemma 6.3 is a family of such linear statistics in t. It is checked on a grid of 24 values of t, with a polynomial lower bound between the grid points. The count of nodes near 1 follows from the field of the positive nodes near $x = 1$, which lies below its supremum by a fixed amount. The window for the fraction of negative nodes follows from polynomial minorants of the indicator functions of the two signs.

6.10. **The trials and their verification.** The concave majorant of §6.7 is bounded at a trial. A trial consists of a reference measure for the free nodes, with a piecewise polynomial density, trial sequences for the rows, the columns and the numerator of Theorem 6.4, and the measure ν' of Lemma 6.6. The trial sequences have 120 Chebyshev coefficients and geometric tails with complex poles, as in (5.3). The bound then reads

$$\begin{aligned}\text{constant} + \alpha \sup(\text{fixed-row fields}) + \sup(\text{free-node fields})\\ + \sup(\text{column field}) \le -U_0 + \tfrac{1}{2}\log 2 + c_2\alpha.\end{aligned}$$

The constant is the value at the trial. It consists of logarithmic energies and moments of the reference measure, the norms of the trial sequences and the entropy of ν'. All of these have closed forms.

The fraction α of fixed columns varies in $[0, \frac{1}{16}]$. We use trials at $\alpha = 0$, $\frac{1}{200}$ and $\frac{1}{16}$. On each of the two intervals between them we interpolate the trials linearly in α, in all the data weighted by mass. Every field is then exactly affine in the interpolation parameter. So its supremum on the interval is bounded by the chord of its suprema at the two ends. The constant is a quadratic polynomial in the parameter. Its curvature is an explicit combination of the four cross energies of the two end trials. The spike constant of Theorem 6.7 interpolates in the same way. At each end trial we therefore bound the suprema of the following fields: the fields of the negative and the positive free nodes, the column field, and the fields of the fixed negative, fixed positive and expanded fixed nodes. Each is a polynomial plus finitely many terms $P(x)\log|x-c|$ and $\operatorname{Re}\rho\log(x-z)$ with complex z, plus a constant.

The suprema are verified on covers by cells, as in §5.6, with the endpoints of the domains treated by Laurent expansions and logarithmic end windows. The polynomial part of a field has degree about 120. On a block of cells we shift its Taylor expansion exactly to order K and bound the remaining coefficients by one table shared by the block. This keeps the integers small.

The three end trials need **??** cells in total. The smallest margin of the budget inequality is 0.037 per unit of N^2.

This completes the proof of the bound of the second case at the rate U_0 for $\alpha \le \frac{1}{16}$. With §5.4 it proves Theorem 2.7.

7. Proof of Theorem 1.1

7.1. **The criterion.** We prove Proposition 2.8. The proof shows where each of the three properties of Q_N enters. The nonvanishing and the smallness are used once, at the end. The divisibility by squares of primes is what makes the choice of the prime free.

Proof of Proposition 2.8. Let a/b be a rational number with b large. Put $L = \log b$, $U = \kappa L$ and $V = (\kappa - \omega)L$, and suppose $|\xi - a/b| \le b^{-\rho\kappa}$. By the prime number theorem, $(1-\delta)t \le \theta(t) \le (1+\delta)t$ for all large t, and $V \to \infty$ with b. If every prime $p \in (V, U]$ had $p^2 \mid b$, then $2(\theta(U) - \theta(V)) \le \log b = L$. But

$$2\big(\theta(U) - \theta(V)\big) \ge 2L\big((1-\delta)\kappa - (1+\delta)(\kappa - \omega)\big) = 2L\big(\omega - \delta(2\kappa - \omega)\big) > L$$

by the fourth hypothesis. So there are primes in $(V, U]$ whose squares do not divide b. Let p be the *largest* one. Every prime $q \in (p, U]$ has $q^2 \mid b$, and $q \le U < 2V < 2p$ because $2\omega < \kappa$. Let B be the product of these q^2. Then $B \mid b$ and $\log B = 2(\theta(U) - \theta(p))$.

Consider the integer $b^{dp} Q_p(a/b) = \sum_i c_i a^i b^{dp-i}$. Each of its terms is divisible by q^{2dp} for each of these q. Indeed q^{2i} divides c_i by hypothesis, and $q^2 \mid b$ gives $q^{2(dp-i)} \mid b^{dp-i}$. So B^{dp} divides the integer. Since $p^2 \nmid b$, $Q_p(a/b) \ne 0$. So $b^{dp} Q_p(a/b)/B^{dp}$ is a nonzero integer, and

$$1 \le \Big(\frac{b}{B}\Big)^{dp} |Q_p(a/b)|. \tag{7.1}$$

Since $\rho p \le \rho U = \rho\kappa L$, we have $|a/b - \xi| \le e^{-\rho p}$. So $|Q_p(a/b)| \le e^{-\eta p^2}$, and (7.1) gives $\eta p^2 \le dp\big(L - 2(\theta(U) - \theta(p))\big)$. With $\theta(U) - \theta(p) \ge U - p - 2\delta U$ this is

$$\eta p \le d\big(L - 2U + 2p + 4\delta U\big).$$

Hence $(\eta - 2d)p \le dL - 2dU + 4d\delta U$. As $\eta < 2d$ and $p \le U$, the left side is at least $(\eta - 2d)U$. So $\eta U \le dL + 4d\delta U$, that is, $\eta\kappa \le d + 4d\delta\kappa$. This contradicts the last hypothesis. □

The largest admissible prime is the right choice. The primes of the window above it have their squares in b, and the divisibility hypothesis recovers these squares from the coefficients. A cruder choice, of any prime in a short interval that does not divide b, loses a factor. The free lower end $V = (\kappa - \omega)L$ allows $\kappa < 2$, because the condition $q \le U < 2p$ for the recycled primes needs only $U < 2V$.

7.2. **The hypotheses for the design D.** Take the design $D = (58, 9, 3, 5, 3)$ and $d = 53$. The polynomials Q_p of the proposition are the Q_N with $N = p$. Let

$$\rho = 296.586094, \quad \eta = 38.33924, \quad \omega = \tfrac{3}{5}, \quad \delta = 2.9 \cdot 10^{-7},$$
$$\kappa = \frac{410}{\rho} = 1.382397\ldots$$

We check the hypotheses.

- $\deg Q_N \le 53N$ by (3.8) and Lemma 3.1.
- The nonvanishing. By Proposition 3.5, $\Delta_p(a/b) \neq 0$ for large p with $p^2 \nmid b$. Hence $Q_p(a/b) \neq 0$.
- The divisibility. By Lemma 3.4, q^{2i} divides the i-th coefficient of Q_p for all primes $p < q \le 2p$.
- The smallness. By Theorem 2.7, Proposition 2.6 and $\log(53N)! = o(N^2)$, for $|x - G| \le e^{-\rho N}$ and large N,

$$|Q_N(x)| \le \exp(-(6431.1338\ldots - 6392.7880\ldots - 3\varepsilon)N^2) \le e^{-\eta N^2},$$

  since $\eta = 38.33924 < 38.34580\ldots$ In the formal proof this is the inequality $\eta < n_0^2(U_0 - \frac{1}{2}\log 2) - \mathsf{A}$ with $\int \kappa$ replaced by its lower bound 22.05325.
- The numerical conditions. $\rho\kappa = 410$ and $0 < \eta < 2d = 106$. Next $\omega = 3/5 \le 1$ and $2\omega = 1.2 < \kappa$. Further $2(\omega - \delta(2\kappa - \omega)) = 1.199998\ldots > 1$. Finally $d + 4d\delta\kappa = 53.0000849\ldots < \eta\kappa = 53.0000857\ldots$

Proposition 2.8 therefore gives $|G - a/b| > b^{-\rho\kappa} = b^{-410}$ for all integers a and all sufficiently large b. This proves Theorem 1.1. The last condition is equivalent to $410 > x = d\rho/(\eta - 4d\delta) = 409.99999\ldots$

8. Remarks on the method

8.1. **The three ingredients.** The exponent $d\rho/\eta$ has three ingredients. The degree $d = n - g$ comes from the construction. Of $\rho = 296.586094$, the energy cost of the fixed columns $n_0 c_2$ is 208.62, and the weight and the binomial factor of one column contribute $\Psi_0 = 87.9663$. Bounding the two separately, by their maxima 52.79 and $58\log 2 = 40.20$, would give $\rho = 301.54258$. The decay rate η is a small difference of two large numbers. The rank deficiencies of Section 4 contribute 22.05 of $\eta = 38.33924$, and the signed bound of Section 6 contributes 8.29824. Since the exponent is inversely proportional to η, a further gain of 0.1 in η lowers it by about 1.1.

8.2. **The determinant of the row functions.** The energy method bounds $|\det \mathsf{L}|$ by 6^n, which is $e^{O(N)}$ and invisible at the scale N^2. At the extremal configuration, numerical evaluation for small N suggests that $\log|\det \mathsf{L}|$ is about $-28N^2$, so most of the true size is not captured. The bound (6.3) loses only $e^{O(N)}$, and Theorem 6.4 is an identity. The loss is in the log-gas bounds of §6.5, which use fixed trial measures, and in the

requirement that the combined functional be bounded by its maximum over all configurations. A sharper treatment of the denominator $Z_s(\psi)$ would raise the gain. These numerical values are not part of the formal proof.

8.3. **Designs.** All the results of Section 4 are lower bounds for the nullity, and the proofs give the kernel vectors explicitly. The closed formulas, the counting inequality and the families derived from it are proved for every design. So are the other arguments of the paper, except the stored subspaces of Section 4.8 and the trials, which belong to the design D. A different design changes d, ρ and η together. We make no claim that D is optimal. Except for one pair of families, the function κ uses no relation between different families. At a prime where windows overlap the exact nullity may be larger than each bound. An upper bound for the nullity is not needed. More independence between families, or larger stored subspaces, would increase η without changing anything else.

9. Formalization

All results of this paper are formalized in Lean 4 [8] with Mathlib [35], on top of OpenAI's formalization of the irrationality of G [26]. The development is available at

`https://github.com/slitsey/catalan-irrationality-measure`.

Theorem 1.1 is the theorem

`CatalanMeasure.GenD.Signed.catalan_irrationalityMeasure_410`

in `CatalanMeasure/GenD/Signed/Data/Assembly/CellsAll.lean`. Its statement uses only Mathlib. Catalan's constant is written as the series $\sum_j (-1)^j/(2j+1)^2$, and the conclusion is $|G - a/b| > b^{-410}$ for all integers a and all sufficiently large b. The modules contain no `sorry` and no `native_decide`. The command `#print axioms` checks that only the three standard axioms `propext`, `Classical.choice` and `Quot.sound` are used. The repository also contains a configuration `comparator.json` for `lake comparator` [21]. It checks that a solution file proves a challenge statement with these axioms only. The prime number theorem is taken from the PNT+ project [36]. The parts of OpenAI's formalization that are used were converted to Lean's module system and ported to a recent Lean release. The repository's `README` and `PORTING.md` describe the changes.

9.1. **Module layout.** The design-generic part of the development is in the directory `GenD` of `CatalanMeasure`. A design is data. The generic modules prove every statement for an arbitrary design, and the instance files contain numerals only.

Generic layer:

`PL` (piecewise linear profiles and their areas), `PrimeGrowth` (the weighted prime number theorem (3.7)), `Budget` (the scalar budgets), `PayW` and `Recycled` (Proposition 2.8), and `Final` (the structure `DesignData` and the theorem `measure_of_design`, which contain no numeral).

Denominators:
`OddProfile`, `TwoAdic`, `DenomExp`, `DenomGrowth`, `Denominator` and `Normalizer`, with `Uniform/Arithmetic` and `Uniform/LedgerW` (the common denominator built from an arbitrary lower bound for the nullity, and the growth of its logarithm).

Rank deficiencies:
`RankT`, `RankU`, `RankU6`, `RankS` and their shared layers for Theorem 2.1; `Uniform/W/` for the counting inequality (`Core`, `Profile`), Lucas's step (`Lucas/PStep`), the families of Section 4.7 and the stored data of Section 4.8 (`Prof/Packet`, `Charts6/Check`); the profile of D with its area `gain532` and the bound `gain532_ge` in `Uniform/Bank532`.

Nonvanishing:
`Nonvanishing`, with the residue-table certificates modulo 1009.

Real place, perturbation:
`Analytic/` (the energy method with fixed points, the column expansion and the quadrature nodes); `BinomVandermonde` (Lemma 5.4) and `Keyhole` (Lemma 5.1) in `GenD/Signed`; `D58/Budget590` (the budgets at the rates U and U_0, and $\rho =$ `rhoEnd`); `Signed/Data/S1` (the 14 cells of the bound for Ψ_0).

Real place, signed bound:
`GenD/Signed`: `PickDet` and `RealPlace` (Lemma 6.1, Theorem 6.2, (6.3)), `LemmaT`, `LemmaTFree` and `HalfDisc` (Lemma 6.3), `RatioIdentity` and `BernsteinSzego` (Theorem 6.4), `TiltedTangent` and `ArcsineDenominator` (Lemmas 6.5 and 6.6), `Spike` and its companions (Theorem 6.7), `ReflectionEnergy` and `ArcsinePhantom` (Theorem 6.8), `MarkedExpansion`, `SigmaCap` and `EnergyNear` (§6.8), `LemmaK` and `Brackets` (§6.9), `Interp` and `Phase2Contract` (the interpolation of trials), and `Gate` and `Theorem410` (the assembly).

Trials and cells:
`Signed/Data/Trial` (the three trials), `Signed/Data/Consts` (the constants of the trials, evaluated in exact arithmetic), `Signed/Data/Cells` (the cell engine and the cells of the three trials) and `Signed/Data/Assembly` (the records of the cell statements and the final theorem).

9.2. **Certificates.** The proof combines conventional formal proofs with numerical certificates checked by the Lean kernel. They are the inverses of the base matrices modulo 1009, the trial sequences and covers of Sections 5 and 6, the cells of the bound for Ψ_0, and the stored subspaces, kernel vectors and window conditions of Section 4.8. All are exact rational or integer computations, decided by the kernel without compiled code. The interval arithmetic used to generate the trial data and the covers [17] is not trusted. Lean rechecks every inequality. Five devices keep the certificates small enough for the kernel.

- *Shared Taylor bounds.* The cells of a cover are grouped in blocks. For each block the derivatives of the polynomial part of the field are evaluated once, exactly, at the centre of the block, together with one bound for the remainder. Every cell of the block obtains its expansion by an exact shift.
- *Split shifts.* In the covers of Section 6, only the low-order coefficients of the shifted expansion are computed exactly. The others are bounded by one crude table per block.
- *Accuracy per term.* The logarithms are evaluated by the series $\log \frac{1+x}{1-x} = 2x \sum_i x^{2i}/(2i+1)$ with an explicit geometric tail. The number of terms is chosen for each logarithmic term separately, from a table of nodes shared by all the cells.
- *Residue recurrences.* The tables of moments modulo 1009 are generated by the one-step recurrences that define them, once proved correct. The kernel checks the recurrence and the products of the matrices with their claimed inverses, row by row, in natural numbers.
- 2*-adic normalisation.* Where a recurrence or a shift produces integers divisible by large powers of 2, the tables store the quotients by these powers, with one lemma relating the two. This also avoids integers that the kernel's hashing of literals treats as equal.

Disclosure of AI use

The mathematics, the Lean formalization, the certificate generators and the text of this paper were produced by AI systems at the request of the author, who posed the problems, directed the work and is responsible for this paper: Claude Opus 5.5, Claude Sonnet 5.5 and Claude Haiku 5.5 (Anthropic), running in Claude Code, and GPT-6.1 Sol (OpenAI), running in Codex. The underlying irrationality proof for G and its Lean formalization are OpenAI's [26].

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